Bank deregulation and personal bankruptcy filings[*]

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Abstract

The increase in economic growth and decline in income inequality after bank deregulation can reduce financial distress. On the other hand, bank deregulation can bring additional risky borrowers to the credit market, which increases the level of financial distress. The results show US bank deregulation -- allowing either a bank to open branches anywhere in the state (intrastate branching) or an out-of-state bank to provide banking services in the state (interstate banking) -- increases Chapter 13 filings in the southeast region. The increase in mortgage defaults and foreclosures in this region after bank deregulation explains the rise in Chapter 13 filings. In contrast to the previous literature, we find an insignificant effect of interstate banking on Chapter 7 filings.

Introduction

During the last quarter of the twentieth century, most states in the US deregulated the banking sector by allowing a bank to open branches anywhere in the state (intrastate bank branching) and by allowing an out-of-state bank to provide banking services within the state (interstate banking). Bank deregulation has increased bank performance, business formations, and economic growth (Jayaratne & Strahan 1998; Black & Strahan 2002). Beck, Levine and Levkov (2010) show that the intrastate branching reduces income inequality by raising the income level of the below-median income group. Tewari (2014) shows that mortgage lending to the lower and middle income groups increases after branch deregulation.[1] These positive outcomes suggest that financial distress should reduce after bank deregulation. On the other hand, bank deregulation allows a bank to spread its geographical coverage and increases its reliance on score based lending, which in turn, allows it to supply credit to a larger customer base. Hence, bank deregulation can bring additional borrowers including those who were earlier rationed out from the credit market (Stiglitz & Weiss 1981). Some of these risky borrowers may not fulfil their debt obligations, which can increase the level of financial distress. These opposite predictions motivate undertaking an empirical study on bank deregulation and financial distress.

The research question of the paper is to assess the effect of the US bank deregulation on personal bankruptcy filings. The main finding is that the Chapter 13 filings increase after deregulation in the southeast region covered by the Sixth Federal Reserve district. In contrast to their decline in the rest of the US, the mortgage defaults and foreclosures in the southeast region increase after the deregulation, which seems to contribute to the rise in Chapter 13 filings. In

[1] Favara and Imbs (2015) show mortgage supply increases after branch deregulation, which in turn, increases the house prices. Kozak and Sosyura (2015) show that stock market participation increases after the interstate banking deregulation.

addition, the replication exercise and the main analysis convince that the documented effect of interstate banking on the Chapter 7 filings is not robust.

The in-depth analysis of the paper suggests its contribution. First, the paper covers both intrastate bank branching deregulation and interstate banking deregulation. Second, it uses aggregated state-level and disaggregated county-level data. Third, it relies on state-of-the-art econometric techniques of parametric and non-parametric difference-in-differences on the sample of all state/county - year observations and on the subsample of contiguous county-pairs of 17 treatment states. To address the data snooping and data mining biases, as a robustness check, the paper obtains the critical values from the placebo sample of counterfactuals. Fourth, it analyzes consumer bankruptcy filings under Chapter 7 (liquidation) and Chapter 13 (reorganization) codes. Finally, the replication exercise as well as the main analysis of the paper contribute in promoting a robust research in financial economics (Harvey 2017).

In the US, a financially distressed borrower can file for personal bankruptcy using either the Chapter 7 or Chapter 13 bankruptcy code. In the case of Chapter 7 bankruptcy — also known as the 'fresh-start' procedure, a borrower's non-exempt assets are liquidated during the bankruptcy proceedings and the resulting amount is used for the payment of unsecured credit. In the case of Chapter 13 bankruptcy, the borrower retains all her personal assets including her home and commits to pay the unsecured creditor over the next three to five years from her future income.[2] Although the secured mortgage lending is not discharged in the bankruptcy process, a financially distressed borrower may file under Chapter 13 code to save her house (White 2006; White & Zhu 2010). Filing for bankruptcy immediately halts the debt collection. This provides additional time to a debtor for arranging funds from friends, relatives, or other sources to pay back mortgage

[2] On average, 70 percent of personal bankruptcy filings are Chapter 7 filings. For more on the bankruptcy chapter choice, see Nelson (1999) and Desai (2016).

arrears. In addition, a bankruptcy judge can request to modify the loan terms, and the mortgage lender may acquiesce. Using 500,000 bankruptcy filings between 1992 to 2005, Dobbie and Song (2015) show that the Chapter 13 bankruptcy protection decreases the five-year foreclosure rate by 19 percentage points. Therefore, given that bank deregulation increases mortgage lending (Tewari 2014), a region with high mortgage defaults and foreclosures witnesses a higher incidence of Chapter 13 bankruptcy filings.

Two schools of thought exist regarding at what level the data need to be aggregated in research involving a state legislation and its impact on the economic outcome. One view is that the aggregated state-level data are appropriate. The rationale is that the independent variable of interest, state bank deregulation, changes across the state lines. Further, the disaggregated county-level data on bankruptcy filings are noisy due to several less-populated counties with negligible bankruptcy filings. The other view is that the aggregation bias can overestimate the effect.[3] First, by aggregating data at the state-level, we lose information on the variation in bankruptcy filings among individual counties. Second, the US counties are not homogenous across states and within a state. A coastal urban county of Virginia is different in terms of economic conditions, demographics, and weather conditions from a rural southwestern county of Virginia. Third, each US state has a different number of counties. Arizona is the sixth largest state in the US in terms of area, however, it has only 15 counties, whereas, Georgia which is ranked 24th in area has 159 counties. In state average analyses, a county of Arizona receives a higher weight than that of a county of Georgia. Finally, the average bankruptcy of a state can be driven by the outlier counties

[3] Holderness (2016) shows usefulness of less-aggregated data for law and finance studies. Using the cross-section of firm-level data, he finds an insignificant relationship between investors' legal protection and ownership structure. This is in contrast to the conventional wisdom, based on the country average data, that the legal protection is inversely related to the ownership structure — the firms of a common (civil) law origin country tend to have less (more) concentrated ownership.

with very low populations. Therefore, to capture the population-weighted effect, one needs to perform a weighted disaggregated regression.

The extant literature provides two difference-in-differences methodologies for analyzing the effect of bank deregulation on bankruptcy filing. First, the traditional national-level study incorporates information on all state/county-year observations (Beck, Levine & Levkov 2010). The ease in sample construction and interpretation of results make this approach popular. In the case of intrastate bank branching, before 1980 (beginning of the sample period of this study), 18 states including District of Columbia already removed the restrictions. By using data of all the states, the branch deregulation effect is underpowered due to small variation in the variable of interest. Further, the national-level study may demonstrate downward bias in estimation due to unobserved heterogeneity in the bankruptcy filings (Dube, Lester & Reich 2010). The alternative approach, as suggested in Huang (2008), is to analyze the state-by-state effect incorporating a sample of contiguous county-pairs.[4] A state line separates the contiguous county pair, and the state on one side deregulates the banking sector in the post-deregulation period, whereas the state on the other side regulates during the entire deregulation window. For this approach, it is essential to have a time difference between the years in which two neighboring states undertake a deregulation event. In the US, 35 states passed legislation allowing interstate banking in a span of four years between 1985 and 1988; therefore, the contiguous county approach is not feasible for assessing the impact of interstate banking. The main advantage of this approach is that it accounts for time-invariant unobserved fixed effects by matching similar counties and provides state-by-state assessment of the deregulation effect.

[4] Holmes (1998), Ljungqvist and Smolyansky (2016), Desai and Elliehausen (in press) are the other studies, to the best of our knowledge, that use contiguous the county approach. Dube, Lester and Reich (2010) use panel data set across contiguous county-pairs and across years to analyze the effect of minimum wage on the employment. The results based on this approach are similar to the ones using all county-year observations in this paper.

The empirical analysis of the paper comprises three parts. In the first part, the results based on all state-year observations for the period 1980-2004 show an insignificant effect of interstate banking on Chapter 7 filings, which is in contrast to the findings of Dick and Lehnert (2010). The replication exercise, as reported in the Appendix A of the paper, shows that the interstate banking effect gains traction when the econometric specification includes unemployment rate as a control variable. And the sample involves state-level data for the period 1980-1994 that includes the states who were late in allowing interstate banking. The first part also includes a dynamic model of deregulation following Beck, Levine and Levkov (2010). An application of this model on the county-level data shows that the Chapter 13 bankruptcy filings gradually increase after branching deregulation.

The second part of the empirical analysis involves a sample of 186 contiguous county-pairs of 17 treatment states. The results show that branch deregulation increases Chapter 13 filings. In addition, the effect of branch deregulation on bankruptcy filings mainly emanates from few states located in the southeast region covered by the Sixth Federal Reserve district. As a robustness check, the non-parametric permutations test, which involves a placebo sample of 385 contiguous county-pairs that were not part of the original treatment sample, confirms that the Chapter 13 filings increase after branching deregulation.

The third part of the empirical analysis concentrates on the deregulation effect in the southeast region. The results show that, in some regressions, Chapter 13 bankruptcy declines in the rest of the US after bank deregulation. However, bank deregulation causes a significant increase in Chapter 13 filings in the Sixth Federal Reserve district. Interestingly, the economic indicators such as unemployment rate, income growth, and income inequality do not substantially differ in the rest of the US and the Sixth Federal Reserve district following bank deregulation. The bankruptcy literature directs us to the mortgage outcomes, for the rise in Chapter 13 filings in the southeast region. Indeed, the results show that negative mortgage outcomes such as less-serious, moderate, and serious mortgage delinquencies, as well as foreclosure starts and foreclosure inventory increase in the Sixth Federal Reserve district after bank deregulation. In the rest of the US, mortgage defaults and foreclosures, actually, decline after bank deregulation. This finding is an ancillary contribution of the paper. Finally, the ratio of the change in Chapter 13 filings to the change in mortgage defaults or foreclosures is positive in the southeast region after bank deregulation, suggesting that delinquent homeowners seek Chapter 13 protection to save their homes.

I. Data

The county-level data on personal (non-business/consumer) bankruptcy filings are from the Report F-5A of the Administrative Office of the U.S. Courts (AOUSC). For the state-level analyses, we add the number of bankruptcy filings of counties of a given state in a given year, and obtain the aggregate data at the state-level. The dependent variables are the Chapter 7 bankruptcy rate and the Chapter 13 bankruptcy rate, which is the number of bankruptcy filings under the Chapter 7 and Chapter 13 procedure in a given year per 1,000 population, respectively. The population data are from the U.S. Census. The sample period is from 1980 to 2004. The main reason of selecting the cut-off year as 2004 is to avoid the influence of the Bankruptcy Abuse Prevention and Consumer Protection Act (BAPCPA) of 2005 and 2007-financial crisis and its aftermath. The county- and state- level bankruptcy data are not available prior to 1980.

The variables of interest are two indicator variables capturing bank deregulation. The indicator variable for interstate banking deregulation takes on a value of one for a given year if a state allows interstate banking in that year, and zero otherwise. The indicator variable for intrastate bank branching deregulation takes on a value of one in a given year if a state permits intrastate bank branching in that year, and zero otherwise. These data are from Huang (2008) and Dick and Lehnert (2010). Following these studies, for Iowa and Hawaii, we take the year of deregulation as 1994 because in that year the Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994 was enacted, which essentially completed the geographic deregulation process. Following the bank deregulation literature, the sample excludes Delaware and South Dakota. These states have consumer finance laws that are favorable to the credit card companies.

The control variables reflect the local economic conditions. These are unemployment rate and income growth. For unemployment rate, we use the value of the previous year and for income growth, we use previous year and current year data. The unemployment rate is the ratio of people seeking a job to the total number of people in the labor market, and is expressed in percentages. The income growth is the change in per capita nominal income relative to the previous year. The income and unemployment data are from the Bureau of Economic Analysis and Bureau of Labor Statistics, respectively.

[Insert Table I here]

Table I reports the summary statistics. Panel A reports the statistics for the state-level data. During the sample period, the average bankruptcy filing under Chapter 7 is 2.274 per 1,000 persons in a typical state. That statistic for Chapter 13 filing is 0.840, suggesting that on average 27 percent of bankruptcy filings are using the Chapter 13 bankruptcy code. Panel B reports the summary statistics using county-level data. The mean and median Chapter 7 bankruptcy rates are 1.914 and 1.529 of a typical county, respectively. These statistics are 0.746 and 0.284 for the Chapter 13 bankruptcy rate.

II. Results using panel data of all state/county-year observations

A. Using all state-level data

Table II reports the results of an analysis using all state-year observations and the following specification.

$$Y_{st} = \alpha + \beta D_{st} + \delta X_{st} + A_s + B_t + \varepsilon_{st} \tag{1}$$

In equation (1), the dependent variable Y denotes the Chapter 7 bankruptcy rate and the Chapter 13 bankruptcy rate. The variables of interest, D, consist of interstate banking and intrastate branching indicator variables. The vector X denotes the time-variant controls. The dummy variables A and B represent state-fixed and year-fixed effects, respectively. The error term is ε, and the subscripts s and t denote state and year, respectively. The robust standard errors are clustered at the state-level (Bertrand, Duflo & Mullainathan 2004). The coefficient β is the difference-in-differences and captures the effect of bank deregulation on the outcome variable Y. The columns (i) to (iv) and (v) to (viii) of Table II have dependent variables Chapter 7 and Chapter 13 bankruptcy rates, respectively.

[Insert Table II here]

The results of Table II show that the local economic conditions explain the variation in Chapter 7 bankruptcy rate. The unemployment rate, however, is insignificantly related to Chapter 13 filing. A possible explanation is that the future income of a borrower plays an important role when the bankruptcy judge decides whether to grant Chapter 13 protection. For a debtor who is out of the labor market, Chapter 7 protection is preferred. More importantly, the interstate banking and intrastate branching deregulations have a statistically insignificant effect on the Chapter 7 and Chapter 13 filings.

The specification of equation (1) shows the average effect of bank deregulation on bankruptcy filings over the sample period. It precludes from analyzing the deregulation effect for each year prior to and posterior to the deregulation event year.[5] It is possible that the impact of state legislation removing geographic banking barriers on its residents' financial distress takes time to materialize. For that, a dynamic model of deregulation, as applied in Beck, Levine and Levkov (2010), is helpful.[6] It is given by the following specification:

$$Y_{st} = \alpha + \beta_1 D_{st}^{-8} + \cdots + \beta_8 D_{st}^{-1} + \beta_9 D_{st}^{+1} + \cdots + \beta_{12} D_{st}^{+12} + A_s + B_t + \varepsilon_{st} \qquad (2)$$

In equation (2), the dummy variable $D^{\mp J}$ takes on a value of one if the year t is J^{th} year prior to or posterior to the deregulation event year of the state s. The negative sign is for the prior year and the positive sign is for the posterior year. The omitted dummy variable, D^0, takes on value of one for the deregulation event year of state s, and zero otherwise. We consider a 20-year span starting from the eighth year prior to and ending on the 12th year posterior to the deregulation year. For the years before the eighth year prior to the deregulation year, the dummy variable D_{st}^{-8} takes on a value of one. Similarly, for the years after the 12th year posterior to the deregulation year, the dummy variable D_{st}^{+12} takes on a value of one. The coefficients, β, on these dummy variables, $D^{\mp J}$, are the parameters of interest. Their significance indicates whether the bankruptcy rate, Y, in a given year differs from that in the deregulation event year. They also show a trend in the deregulation effect. A figure better explains the results of equation (2) than a table.

[Insert Figure 1 and Figure 2 here]

[5] The deregulation event year is the year in which a state enacts a legislation removing the regulation.

[6] We sincerely thank Ross Levine for posting the data and STATA® codes of their paper on his website. Autor (2003) and Tewari (2014) also use this dynamic model for their analyses.

Figure 1 plots the deregulation dynamics on the Chapter 7 bankruptcy rate. The left plot shows the innovations relative to the interstate banking deregulation, and the right plot shows those with respect to the intrastate branching deregulation. The black dots are the magnitude of the coefficients on dummy variables, $D^{\mp J}$, and the dash vertical spikes refer to the 95% confidence interval. The effect is significant if the vertical spike fails to cross the horizontal line passing from zero. As shown in the left plot of Figure 1, the difference of the average Chapter 7 bankruptcy rate in a given year and that in the deregulation year increases after a state permits interstate banking. However, the difference is not statistically significant for any given year.

Figure 2 plots the dynamics of the deregulation effects on the Chapter 13 bankruptcy rate. As shown in the left plot, for the years prior to the interstate banking deregulation, the average Chapter 13 bankruptcy rate is close to that for the deregulation year. As the interstate banking deregulation kicks in, the average Chapter 13 bankruptcy rate gradually increases relative to its level in the deregulation year. The intrastate branching effect on Chapter 13 filings, as shown in the right plot, looks interesting. Prior to the branching deregulation, the average Chapter 13 bankruptcy rate is higher than that in the year of branching deregulation. The difference, however, gradually drops as we approach closer to the deregulation event. Then, the trend reverses. Once the branching deregulation is completed, the Chapter 13 bankruptcy rate in a given year relative to its value in the branching deregulation year steadily increases. Overall, the two plots of Figure 2 suggest that the Chapter 13 filings might have increased post-banking deregulation. The effect is not statistically significant due to less variation in the state-level data, and perhaps, the county-level data may provide more information.

B. Using all county-level data

This subsection reports the results using the disaggregated county-level data. The regressions use the following linear specification.

$$Y_{cst} = \alpha + \beta D_{st} + \delta X_{cst} + A_c + B_t + \varepsilon_{cst} \quad (3)$$

In equation (3), the subscript c denotes a county. The dependent variables, Y_{cst}, are the Chapter 7 and Chapter 13 bankruptcy rates of county c of state s in a given year t. The vector, X_{cst}, is for the time-variant controls of a county. The vectors A_c and B_t are the dummy variables controlling for county and year fixed effects, respectively. The remaining description of equation (3) is same as that of equation (1). The standard errors are clustered at the state-level. All regressions are weighted by the average population of a county during the sample period.[7]

[Insert Table III here]

Table III reports the results using all county-year observations. The dependent variables for the columns (i) to (iv) and (v) to (viii) are Chapter 7 bankruptcy rate and Chapter 13 bankruptcy rate, respectively. As tabulated in Table III, the effect of local economic conditions on bankruptcy filings persist. The income growth decreases bankruptcy filings under both codes and the unemployment rate increases Chapter 7 filings. The coefficients of the indicator variables measuring bank deregulation are statistically indistinguishable from zero for Chapter 7 and Chapter 13 bankruptcy rates. The magnitude of the intrastate branching effect on Chapter 13 filings is between 0.088 and 0.104. The average value of the Chapter 13 bankruptcy rate in a typical county before the branching deregulation is 0.195. This statistic is untabulated. Therefore, the intrastate branching effect on Chapter 13 filings seems to have an economic significance.

Next, the following regression model helps analyze the dynamics of the deregulation effect using the county-level data.

[7] In their cross-sectional analyses involving US counties data, Mian and Sufi (2014) use population-weighted regressions. We thank Amir Sufi for providing the replication kit of their study on his website.

$$Y_{cst} = \alpha + \beta_1 D_{st}^{-8} + \cdots + \beta_8 D_{st}^{-1} + \beta_9 D_{st}^{+1} + \cdots + \beta_{12} D_{st}^{+12} + A_c + B_t + \varepsilon_{cst} \tag{4}$$

The notations for equation (4) are similar to those for equation (2), only addition is the notation c for a county. The results of a dynamic deregulation model using county-level data are shown in Figure 3 and Figure 4.

[Insert Figure 3 and Figure 4 here]

Figure 3 shows the dynamics of bank deregulation effects on Chapter 7 filings based on US county-level data. As shown in the left plot, the level of Chapter 7 bankruptcy filings does not change substantially as we approach closer to the interstate banking year. After the interstate banking deregulation, the Chapter 7 filings decrease relative to its level during the deregulation year. A similar trend is observed for the intrastate branching deregulation, as shown in the right plot. However, each innovation is statistically indistinguishable from zero.

Figure 4 plots the difference in the average Chapter 13 filings in a given year relative to that in the deregulation year using county-level data. The removal of restrictions on interstate banking, as shown in the left plot, seems to have no impact on the Chapter 13 bankruptcy rate using all county-year observations. The innovations, for the most years, are close to zero. The plot on the right-hand side is promising. The Chapter 13 filing in a typical county for a year prior to branching deregulation is almost same for that in a deregulation year for that county. However, after branch deregulation, the Chapter 13 filing increases at a notable pace. For some years, the 95% confidence intervals, as shown by the vertical spikes, are close to the horizontal line passing from zero, suggesting a modest statistical significance. More importantly, a comparison of the pattern of our plot to that of Figure 1 in Tewari (2014) provides a direction. In the latter plot, the author shows that the mortgage lending consistently increases after intrastate branching. Therefore, a link among intrastate branching, mortgage credit, and Chapter 13 filings seems to exist. Perhaps, a set of research tools that allow us to scratch beneath the surface can help.

III. Branch deregulation and bankruptcy filings in the contiguous county-pairs

This section documents the results using the sample of contiguous county-pairs. The research design precludes an analysis on interstate banking deregulation. There are five steps in this approach. The first step is to form a treatment sample of contiguous county-pairs. In the second step, a visual inspection helps satisfy the common trend assumption. The third step involves linear regressions on the treatment sample to ascertain the branch deregulation effect. The fourth step is a variant of the third step, the only difference is that we use subsamples of individual treatment states. Finally, to address the data mining and data snooping biases, the fifth step conducts a randomization (permutations) test involving a placebo sample and counterfactuals.

A. Treatment sample and contiguous county-pairs

The research design involves identifying a border segment and contiguous county-pairs along that segment.[8] A border segment separates a treatment and its control state. In the branch deregulation (event) year, a treatment state removes restrictions on intrastate bank branching. The event window consists of the pre-event period, event year, and the post-event period. Throughout the event window, the control state prohibits intrastate bank branching. Therefore, a contiguous county-pair has a treatment county and a control, and the state border separates them. As an example, the border-segment of Georgia and Florida is in the treatment sample. Georgia permitted bank branching in 1983, and five years later, Florida did the same. The event year is 1983, Georgia

[8] We follow Huang (2008) to construct the treatment sample. We also thank Rocco Huang for providing a list of contiguous county-pairs on his website.

is a treatment state, and Florida is a control state. Grady county of Georgia and Leon county of Florida is a contiguous county-pair.

The rationale for matching based on geographic proximity is that the economic, demographic, cultural and weather conditions are similar in the treatment and control counties. By focusing on the difference-in-differences in bankruptcy filings, we account for the heterogeneity in the unobservable fixed effects. A statistically significant *net* change in the bankruptcy filings in a treatment state indicates that the effect is due to the intrastate bank branching in the treatment state. Since it is important to have a match pair of similar counties, following Huang (2008), the treatment sample considers counties of the eastern US.

[Insert Table IV and Figure 5 here]

Table IV reports the border segments and their number of contiguous county-pairs of the treatment sample. Between 1981 and 1990, seventeen US states qualify as a treatment state, and among them there are 23 border segments and 186 contiguous county-pairs. Alabama, Tennessee, and Missouri have the highest number (27, 25, and 28, respectively) of contiguous county-pairs. The lowest numbers of county-pairs are for Texas, Massachusetts, and West Virginia (2, 3, and 4, respectively). A state can be a treatment and/or control state depending on its branching deregulation year and that of its neighboring state. As an example, Missouri allowed intrastate bank branching in 1990. Its border segments with Nebraska, Tennessee and Kansas make it a control state, whereas, its border segments with Arkansas and Iowa make it a treatment state. Figure 5 shows the contiguous county-pairs on the US map. The treatment counties are in green and their comparison [control] counties are in red.

The pre-event period is defined as the three years prior to branching deregulation. However, due to data limitation this period may be shorter, but not less than one year.[9] The post-event period is defined as three years after the deregulation year. However, it is two years in some cases due to the subsequent removal of branching restrictions in the control state.[10] By requiring a minimum of two years for the post-period, a reasonable amount of time is given for the branch deregulation to show its effect on bankruptcy filings. It also helps avoid overlapping of the post-event period of the treatment state and the year in which the control state allows bank branching. In all analyses, the branch deregulation year is excluded in both pre- and post-periods.

The three-year length of the post-period can raise a genuine concern. The intrastate branching deregulation and consumer bankruptcy outcomes are likely to take time to diffuse. Once branching deregulation comes into effect, it takes time for the market structure to evolve, and furthermore, there is a time lag between consumers taking out debt and the increased debt leading to bankruptcy. Unfortunately, the timing of branching deregulation in the US states and the research design preclude extending the post-event period beyond three years. In most border-segments, after the third year of the post-event period, the actual event window of the control state begins. Therefore, the meaningful comparison of treatment and control counties is difficult beyond the third year due to the overlapping of event windows. On a positive note, the observed deregulation effect, if any, is one of the most conservative estimates of the true effect.

B. Descriptive statistics of bankruptcy rates for treatment and control samples

[9] As examples, in the case of Alabama, Pennsylvania, Georgia, the pre-event period is a minimum of one year (1980), two years (1980 and 1981), and a maximum of three years (1980, 1981, and 1982), respectively.

[10] As an example, in the case of the Massachusetts (treatment)-New Hampshire (control) border segment, the post-event period is two years (1985 and 1986).

With the chosen criteria of pre- and post- event periods, the number of contiguous county-pair and year observations is 1,207. Out of these, for five observations, the bankruptcy filings of treatment counties are missing. Therefore, the sample is 1,202 x 2 = 2,404 county-year observations.

[Insert Table V here]

Table V reports the descriptive statistics of bankruptcy filings. These statistics are on two dimensions — year relative to the branching deregulation and whether a county is in the treatment or control group. The reported values are mean, standard deviation, and the number of counties. For example, one year prior to the branch deregulation year, the average values of Chapter 7 and Chapter 13 bankruptcy rates are 0.966 and 0.178 for the sample of control counties, and 0.9657 and 0.286 for the sample of treatment counties. The average Chapter 7 and Chapter 13 bankruptcy rates increase over time, suggesting a time trend in bankruptcy filings. The bottom part of Table V reports the summary statistics for the pre-deregulation and post-deregulation periods. The average Chapter 13 bankruptcy rates are 0.154 and 0.263 for the control and treatment groups during the pre-period. In the post-period, these statistics are 0.266 and 0.524 for the control and treatment groups, respectively. Thus, the *net* increase in Chapter 13 filings in the treatment group due to the intrastate bank branching is 0.149 [≈ (0.524 - 0.263) - (0.266 - 0.154)]. This deregulation effect has an economic significance, because the average Chapter 13 bankruptcy rate is 0.263 in the pre-period for the treatment group.

C. Parallel (Common) trend assumption

The underlying assumption is that the time trends in bankruptcy filings among the treatment and control groups remain the same in the absence of branch deregulation. This common trend assumption does not imply that the average values of the bankruptcy filings in both groups are the same. Therefore, the bankruptcy evolution for both groups should be parallel prior to branching deregulation, and start to diverge (but probably not sharply) afterward. It is difficult to test the common trend assumption; however, one can look at the pre-period bankruptcy filings to show that the trends in both groups are moreover similar.[11]

[Insert Figure 6 here]

Figure 6 shows trends in Chapter 7 filings and Chapter 13 filings for treatment and control groups relative to the year in which the treatment group allowed intrastate bank branching. The data are not available for all the counties for all prior years due to data unavailability before 1980. The two short-dash lines show the pre-event period, the long-dash line shows the intrastate bank branching deregulation year, and two dash-dot lines show the post-event period. Throughout the period [-8, +3], the average Chapter 7 bankruptcy filing in the treatment sample is more than that for the control sample. The trends look similar before the beginning of the pre-event period. During the pre-event period, the difference in average Chapter 7 bankruptcy starts declining. However, the divergence in the Chapter 7 filings path for both groups is noticeable in the third year after the event year.

The trend of Chapter 13 filings for the treatment group is gradually increasing, however, that increase is steep in the post-event period. The trend in Chapter 13 filings for the control group is moreover flat till the event year, and then increases marginally during the post-event period.

[11] For more on the common trend assumption in the law and finance literature, see Waldinger (2016), Bhutta, Goldin and Homonoff (in press), and Desai and Elliehausen (in press), among others.

During the pre-event period, the trends in Chapter 13 filings for treatment and control groups are parallel, and they diverge significantly during the post-event period.

Overall, the Figure 6 plots suggest that Chapter 7 and Chapter 13 filings in treatment and control samples seem to satisfy the common trend assumption. In addition, the bottom graph indicates that the Chapter 13 bankruptcy rate increases after the intrastate branching deregulation.

D. Regression-based difference-in-differences involving contiguous county-pairs sample

This subsection reports the results of regressions on the treatment sample of all 186 contiguous county-pairs. The following empirical specification is used to assess the effect of intrastate bank branching on personal bankruptcy filings.

$$Y_{i,t} = \beta_0 + \beta_1 \times Treat_i + \beta_2 \times Post_t + \delta \times (Treat * Post)_{i,t} + \gamma \times Z_{i,t} + \varepsilon_{i,t}\ , \qquad (5)$$

where the dependent variable *Y* is for the Chapter 7 bankruptcy rate or the Chapter 13 bankruptcy rate. The subscripts i and t index for the county and year. The treatment indicator variable (*Treat*) takes on a value of one for a treatment county and zero for a control county. The post-period indicator variable (*Post*) takes on a value of one if the year *t* is in the post-event period and zero if it is in the pre-event period. The vector *Z* reflects the county-level economic conditions. It includes the previous year unemployment rate, and previous and current nominal income growth. The variable of interest is the interaction term of the treatment and post-period indicator variables. The coefficient δ indicates the *net* change in bankruptcy filing for the treatment group after its branching deregulation. A positive value indicates the bankruptcy rate has increased after the intrastate bank branching deregulation. In all regressions, robust standard errors are used to control for possible heteroscedasticity. The branch deregulation event year is excluded, therefore the sample size is 2,034.

The data set is not the traditional panel data set of counties for the sample period of 1980 to 2004 (Table III). The time component is measured relative to the branching deregulation year. Further, since the unit of analysis is a contiguous county-pair of a border segment, some states are considered both treatment and controls depending on which border segment is under consideration. Therefore, the robust standard errors are not clustered at the state-level.

[Insert Table VI here]

The dependent variables for the columns (i) to (iii) and (iv) to (vi) of Table VI are Chapter 7 bankruptcy rate and Chapter 13 bankruptcy rate, respectively. The columns (i) and (iv) are the base-case (univariate) model. As shown in column (iv) of Table VI, the coefficient on constant term is 0.154. It is the average Chapter 13 bankruptcy rate for the control group in the pre-event period. The sum of the coefficients on the constant and post-period indicator is 0.154 + 0.112 = 0.266; it is the average Chapter 13 bankruptcy rate for the control group in the post-event period. The sum of the coefficients on the constant term and the treatment indicator is 0.263, which is the average Chapter 13 bankruptcy rate in the pre-event period for the treatment group. The sum of the coefficients on the constant, post-period indicator, treatment indicator, and the interaction of two indicator variables is 0.524. It is the average of Chapter 13 filings for the treatment group in the post-event period. The difference-in-differences is 0.149, it is the net change in average Chapter 13 bankruptcy rate of the treatment group and that of the control group. All these numbers are the same as those mentioned earlier during the description of Table V. This net increase in Chapter 13 filings is statistically significant at the 1 percent level.

In summary, the results of Table VI show that the Chapter 13 filings increase following the intrastate branching deregulation, while the branching effect on Chapter 7 bankruptcy filings is insignificant.

E. Individual state-level analyses using the contiguous county-pairs sample

This subsection reports the branching deregulation effects for each of the 17 treatment states. The individual state-level analysis helps assess the heterogeneity in the deregulation effect across states. More specifically, we run regressions using equation (5) on the subsample of a treatment state. To economize on space, Table VII reports only the coefficients on the interaction term capturing the difference-in-differences. The columns (i) to (iii) and (iv) to (vi) have dependent variables Chapter 7 and Chapter 13 bankruptcy rates, respectively. The independent variables in the specifications are the same as those in the Table VI. The sum of all observations in the last column equals to 2,034, which was the sample size for the results of Table VI.

[Insert Table VII here]

In the case of Tennessee, the contiguous county-pairs are 25. From these pairs, the sample size for the regressions is 25 x 2 x 6 = 300. As per the base specification (iv), the branching effect on Chapter 13 filings is 0.752, and it is statistically significant at the 1 percent level. The average Chapter 13 filing during the pre-event period is 0.765 for the Tennessee treatment sample.[12] Therefore, the net effect is highly economically significant. In summary, the results as reported in Table VII suggest that the intrastate bank branching effect is heterogeneous across states and type of bankruptcy code. After the removal of bank branching restrictions, the Chapter 7 filings, on average, increase in Alabama, Georgia, and Mississippi. The Chapter 13 filings, on average, increase in Tennessee and Louisiana and decrease in Missouri after the intrastate bank branching.

[12] It is reported in the Appendix Table B2.

F. Non-parametric test based on counterfactuals

The branching deregulation effect on Chapter 13 filings, as reported in Table VI, may be by chance. The intrastate bank branching event is not randomly selected. The data set on personal bankruptcy filing is common for all researchers. In addition, there is an upward time trend in bankruptcy filings during the branch deregulation period (1980-1994) of the study. Therefore, it is possible that the findings are due to data snooping and data mining. One approach to address this criticism, as suggested in Huang (2008), is to conduct a randomization (permutations) test, which is used in clinical trials. The following five steps can summarize this non-parametric test. First, we identify border segments of those 17 treatment states that were *not part* of the original treatment sample. The second step is to form a placebo sample of contiguous county-pairs on these non-event border segments. In the third step, using the placebo sample, we construct a data set of counterfactuals of fictitious treatment years and contiguous county-pairs. The selection criteria of a fictitious treatment year are that it should not be part of the actual treatment windows of the treatment and its control states, and it is within the branch deregulation period. Using the counterfactuals, we construct a distribution of the fictitious treatment effect in the fourth step. Finally, the critical values of this distribution enable us to assess the branch deregulation effect on the actual treatment sample.

The details of the non-parametric test are in Appendix B. The findings of the nonparametric test are similar to those of the parametric test reported in earlier subsections. The branching deregulation increases the Chapter 13 filings, and the effect is statistically significant at the 1 percent level. In addition, the intrastate branching deregulation increases the Chapter 13 filings in Tennessee and Louisiana, and it increases Chapter 7 filings in Alabama, Georgia, and Mississippi.

In summary, the results of Section III suggest two points. First, the intrastate branching deregulation increases the Chapter 13 filings. Second, the effect of branching deregulation on bankruptcy filings is limited to few states. These are Alabama, Georgia, Mississippi, Tennessee, and Louisiana. Interestingly, all these states are in the southeast region, which is covered by the Sixth Federal Reserve district — the Atlanta Fed region.[13]

IV. Bank deregulation, bankruptcy filings, and the Sixth Federal Reserve District

This subsection consists of three parts. The first part compares the bankruptcy filings in the Sixth Federal Reserve district with those in the rest of the US, before and after the bank deregulation. The second part compares the economic indicators and mortgage outcomes in the Sixth Federal Reserve district with those in the rest of the US, before and after the bank deregulation. Finally, the third part analyzes the deregulation effect in the Sixth Federal Reserve district after controlling for mortgage outcomes.

A. Deregulation effect on bankruptcy filings in the Sixth Federal Reserve District

Figure 7 shows average bankruptcy filings for the Sixth Federal Reserve district and the rest of the US relative to the bank deregulation year. Panel A uses the state-level data, whereas the Panel B uses the county-level data.[14] Each panel contains four plots, showing Chapter 7 and Chapter 13 filings relative to the interstate banking and intrastate branching deregulations.

[13] Appendix C reports an analysis checking the parallel trend assumption for selected states of the southeast region.

[14] The Sixth Federal Reserve district covers all counties of AL, GA, and FL, 74 counties in the eastern two-thirds of TN, 38 counties in southern LA, and 43 counties in southern MS. For the state-level analysis, we include TN, LA, and MS in the Sixth Federal Reserve district.

[Insert Figure 7 here]

As shown in Figure 7, there is no significant difference in the deregulation effect in the Sixth Federal Reserve district and that in the rest of the US, as far as the Chapter 7 filings are concerned. That is not the case with the Chapter 13 bankruptcy filings. The average Chapter 13 bankruptcy rate in the Sixth Federal Reserve district is higher than that for the rest of the US for most years relative to the bank deregulation year. Prior to the bank deregulation, the Chapter 13 bankruptcy paths are almost parallel, except during the few years prior to the deregulation. The paths diverge significantly after the bank deregulation. The increase of Chapter 13 bankruptcy filings in the Sixth Federal Reserve district outpaces that in the rest of the US. These patterns are similar for the deregulation type and the data aggregation.

A regression analysis formalizes the observations of the above plots. Further, it helps control for the time trends in bankruptcy filings, state-specific effects, and local economic factors. A modified version of the equations (1) and (3) captures the bank deregulation effect in the southeast region. It includes an interaction term of an indicator variable for bank deregulation and that for the Sixth Federal Reserve district. For the state-level data, it is given by the following linear equation:

$$Y_{st} = \alpha + \beta D_{st} + \gamma(D_{st} * Sixth_s) + \delta X_{st} + A_s + B_t + \varepsilon_{st}, \tag{6}$$

where the indicator variable $Sixth_s$ takes on a value of one if the state s is AL, FL, GA, TN, MS, or LA, and zero otherwise. The coefficient of interest is γ on the interaction term, its positive value indicates that the bank deregulation effect is higher in the Sixth Federal Reserve district than that for the rest of the US. The equation (6) excludes an indicator variable for the Sixth Federal Reserve district, because it includes all state dummies.

[Insert Table VIII here]

Panel A of Table VIII reports the results using the state-level data. The dependent variables are the Chapter 7 bankruptcy rate and the Chapter 13 bankruptcy rate for the columns (i) to (iv) and columns (v) to (viii), respectively. The coefficient on the interaction term of bank deregulation and the Sixth Federal Reserve district is statistically indistinguishable from zero for the Chapter 7 bankruptcy rate. Therefore, the change in Chapter 7 bankruptcy rate after the bank deregulation in the southeast region does not significantly differ from that in the rest of the US. That is not the case for the Chapter 13 filings. As shown in the column (v), the coefficient on intrastate branching indicator is -0.232. It shows that the average Chapter 13 bankruptcy rate in the rest of the US has declined by 0.232 after the intrastate branch deregulation, and the decrease is statistically significant at the 10 percent level. This demonstrates a positive effect of the branch deregulation in reducing financial distress. However, as shown by the coefficient on the interaction term of branch deregulation and the Sixth Federal Reserve district, the average Chapter 13 bankruptcy rate in the Sixth Federal Reserve district has increased by 1.244 after the removal of intrastate bank branching restrictions in that region. The change is statistically significant at the 1 percent level. The interstate banking deregulation has also increased the average Chapter 13 bankruptcy rate in the southeast region, as shown in the columns (vii) and (viii).

Panel B of Table VIII reports the results using county-level data. One of the advantages of using county-level data is that the indicator variable for the Sixth Federal Reserve district appropriately covers the actual area covered by the Federal Reserve Bank of Atlanta. The following linear regression is estimated:

$$Y_{cst} = \alpha + \beta D_{st} + \gamma(D_{st} * Sixth_c) + \delta X_{cst} + A_c + B_t + \varepsilon_{cst} \qquad (7)$$

In equation (7) , the indicator variable $Sixth_c$ takes on a value of one for all counties of AL, GA, and FL, 74 counties in the eastern two-thirds of TN, 38 counties in southern LA, and 43 counties in southern MS. The results show that the interstate banking deregulation has increased the Chapter 13 bankruptcy filings in a typical county of the southeast region.

Overall, the results of Table VIII and the plots of Figure 7 show that the Chapter 13 bankruptcy filings increase in the Sixth Federal Reserve district after the bank deregulation. In the remaining of the paper, we shall seek the explanations for this rise.

B. Deregulation effect on economic indicators in the Sixth Federal Reserve District

The most intuitive candidates to explain the rise in the Chapter 13 filings in the Sixth Federal Reserve district after the bank deregulation are the local economic indicators. These are the income inequality, unemployment rate, and income growth. The hypothesis is that the economic conditions in the southeast region have worsened after the bank deregulation, therefore, the level of financial distress is increased in that region, and the borrowers are filing Chapter 13 to recover from the indebtedness.

The additional variables required for the analysis are pertaining to the income inequality. These are the Gini coefficient and the Theil index. The Gini coefficient is derived from the Lorenz curve. If all individuals of a state receive the same level of income in a given year, then the value of the Gini coefficient is equal to zero, suggesting no income inequality. On the other hand, if all state income is earned by one individual, then the Gini coefficient is equal to one, suggesting the highest level of income inequality. The other measure is Theil index. It is equal to one if all individuals get the same income, and it is equal to natural logarithm of N if only one individual derives all income, where N is the number of individuals. Beck, Levine and Levkov (2010) provide a detailed description of these measures. We obtain the necessary data from Ross Levine's website, and report the descriptive statistics in Appendix Table D1 of the paper.

Table IX reports the bank deregulation effects on local economic indicators for the Sixth Federal Reserve district and for the rest of the US. The econometric specification is the same as that in equation (6), except the outcome variable captures the local economic condition instead of bankruptcy filings, and the time-variant controls, X_{st}, are excluded. The columns (i) to (iv) have dependent variables measuring the income inequality, and columns (v) to (viii) have dependent variables for unemployment and income growth.

[Insert Table IX here]

As shown in the Table IX, the income inequality in the rest of the US has decreased after both intrastate branching and interstate banking deregulations. The unemployment rate has declined and the economic (income) growth has increased after the branch deregulations in the rest of the US. These findings show the positive effects of bank deregulation in the rest of the US and confirm the previous findings of Beck, Levine and Levkov (2010) and Jayaratne and Strahan (1996). More importantly for our research, the coefficients on the interaction terms, in all but one of the regressions, are statistically indistinguishable from zero. These findings illustrate no discernible change in the economic conditions in the Sixth Federal Reserve district after the bank deregulation, in comparison with that in the rest of the US. Therefore, we need to focus on the factors other than the economic indicators to explain the rise in Chapter 13 filings post-deregulation.

The finding that the income inequality in the Sixth Federal Reserve district has further reduced after the interstate banking deregulation, as shown in column (ii) of Table IX, can guide us. A linkage of this finding with the results of Tewari (2014) and the conjecture of White (2006) indicates focusing on mortgage outcomes. An improvement in the income inequality might have increased mortgage lending in the Sixth Federal Reserve district. Lending to some of the additional homeowners may be risky, resulting in the increase in negative mortgage outcomes such as defaults and foreclosures. The US bankruptcy code under Chapter 13 is designed to protect a borrower's assets including her home. Therefore, the hypothesis is that the negative mortgage outcomes in the southeast region have increased, whereas in the rest of the US they have decreased, after the bank deregulation.

C. Deregulation effect on mortgage outcomes in the Sixth Federal Reserve District

The state-level data on mortgage outcomes for the period 1980 to 2004 are from the National Delinquency Survey.[15] Two forms of mortgage outcomes are considered on non-FHA non-VA residential loans — mortgage delinquencies and foreclosures. The three types of mortgage delinquencies are the percentages of mortgages for which the payments are due above 30 days, above 60 days, and above 90 days. A mortgage is in default if the payment is due above 90 days. The mortgage delinquencies exclude the mortgages that are in the foreclosures. The two measures of foreclosures are 'foreclosure started' and 'foreclosure inventory'. The foreclosure started represents the percentage of mortgages that entered the foreclosure process during a given year, and it is a flow measure. The foreclosure inventory represents the percentage of mortgages in the foreclosure process at the end of the year, and it is a stock measure. Appendix Table D1 reports the summary statistics of the mortgage outcomes. Out of these five mortgage outcome measures, the most relevant for our study are the payments due above 90 days and the foreclosure started. A homeowner who is in serious delinquency (default) on her mortgage payments or whose house has

[15] It is a quarterly survey on residential mortgages released by the Mortgage Bankers Association. The survey participants include over 120 mortgage lenders including mortgage banks, commercial banks, thrifts, S&Ls, subservicers, and life insurance firms. For more information on this survey, see Desai, Elliehausen and Steinbuks (2013).

just entered into the foreclosure process is more likely to seek protection under the Chapter 13 bankruptcy code.

[Insert Figure 8 here]

Figure 8 shows average mortgage defaults and foreclosure starts for the Sixth Federal Reserve district and that for the rest of the US relative to the bank deregulation year. As shown in the upper two plots, before the bank deregulation, the mortgage default rate in the rest of the US is higher than that in the southeast region. After the deregulation, the situation is reversed. The mortgage defaults in the rest of the US gradually decrease after both forms of bank deregulations. However, the trends are opposite for the Sixth Federal Reserve district. Further, as shown in the bottom two plots, the foreclosure start rate increases in the Sixth Federal Reserve district, for at least five years, after both forms of bank deregulation. On the other hand, for the rest of the US during this period, it declines after the intrastate branching and remains almost flat after the interstate banking.

[Insert Table X here]

Table X reports the results analyzing mortgage outcomes. The empirical specification is based on equation (6). For completeness, all five mortgage outcomes are included as dependent variables, in turn. Panel A and Panel B show the results for the intrastate branch deregulation and interstate banking deregulation, respectively. Within these panels, there are three sub-panels. The first sub-panel includes only the deregulation indicator as an independent variable. The second sub-panel adds the interaction term of the deregulation indicator and the Sixth Federal Reserve district indicator. Finally, the third sub-panel adds economic indicators as controls. All the regressions include year and state dummies, and the robust standard errors are clustered at the state-level.

Panel A1 of Table X shows that the mortgage defaults and foreclosure starts in the US have decreased after the intrastate branching deregulation, suggesting a positive outcome of the branch deregulation. In addition, Panel A2 and Panel A3 show that the mortgage defaults have decreased after branch deregulation in the US excluding the southeast region. These ancillary findings, by themselves, are of interest. To the best of our knowledge, Tewari (2014) is the only study to report an insignificant effect of the branching on the mortgage defaults.[16] The statistical insignificance may be due to the unusual default behavior in the southeast region post-deregulation. In all the regressions of Table X, the coefficient on the interaction term of branch deregulation and the southeast region is positive and statistically significant at the 1% level. As an example, the coefficient on intrastate branching is -0.176. It shows that, after the branch deregulation, the average mortgage default has declined by 0.176 percentage-points in the rest of the US. This change by itself is statistically significant at the 5% level. The coefficient on the interaction term is 0.568. It shows that, after the branch deregulation, the average mortgage default in the Sixth Federal Reserve district has increased by 0.568 percentage-points, which is significant at the 1% level.

In summary, the results documented in Table X and the plots of Figure 8 provide two findings. The mortgage defaults and foreclosure rates in the Sixth Federal Reserve district have increased after removal of the geographic restrictions on the banking activities. In addition, in the rest of the US, we find some evidence of the decline in negative mortgage outcomes after the bank deregulation, suggesting the benefits of the bank deregulation. Together, these findings help provide an explanation for the rise in the Chapter 13 filings in the southeast region.[17]

[16] See footnote 14 of Tewari (2014).

[17] Appendix Figure E1 shows the graphs for the FHA and VA loans. The regression results are robust to the FHA and VA loans.

D. Mortgage outcomes and Chapter 13 filings in the Sixth Federal Reserve District

This subsection reports the effects of mortgage defaults and foreclosure starts on Chapter 13 bankruptcy filings with a focus on the Sixth Federal Reserve district. The main objectives are to ascertain (1) whether the bank deregulation increases Chapter 13 filings in the southeast region after controlling for mortgage outcomes, and (2) whether an increase in mortgage default or foreclosure can cause an increase in Chapter 13 filings in the Sixth Federal Reserve district after the bank deregulation. The former helps confirm that the bank deregulation has indeed increased the Chapter 13 filings in the southeast region, even after controlling for its higher level of mortgage defaults and foreclosures. The latter helps to test whether Chapter 13 filings, indeed, help save the home, using bank deregulation in the southeast region as a quasi-experimental setting.

[Insert Table XI here]

Table XI reports the results assessing the bank deregulation effect on Chapter 13 filings in the Sixth Federal Reserve district after controlling for mortgage defaults and foreclosure starts. The linear regression model is a modified version of equation (6) with mortgage defaults and foreclosure started as additional time-varying control factors. The columns (i) to (iv) and (v) to (viii) have payment due above 90 days and foreclosure started as additional factors, respectively. The results show that the mortgage defaults and foreclosures started increase the Chapter 13 filings. This supports the conjecture that the Chapter 13 filings help save the home (White 2006). More importantly for our study, the interaction term on bank deregulation and the Sixth Federal Reserve district remains significant. This result suggests that after controlling for mortgage defaults and foreclosures still the bank deregulation increases Chapter 13 filings in the Sixth Federal Reserve district.

In the final analysis, our objective is to check whether the increase in mortgage defaults or foreclosure starts leads to an increase in Chapter 13 bankruptcy filings in the Sixth Federal Reserve district after the bank deregulation. Therefore, the empirical model needs a triple interaction term involving two indicator variables — bank deregulation and the Sixth Federal Reserve district, and a continuous variable — either payment due above 90 days or foreclosure started. For model completeness, we also include the combinations of the triple interaction term components.

[Insert Table XII here]

Panel A of Table XII reports the results for intrastate branching deregulation. Within this panel, the subpanel A1 includes payments due above 90 days, whereas the subpanel A2 includes foreclosure started. As shown in the Panel A, there is some evidence that the increase in mortgage default or foreclosure start causes an increase in Chapter 13 filings for the Sixth Federal Reserve district after its branching deregulation. These effects, however, are observed in four of the six regressions with the statistical significance of 10% level. Panel B of Table XII reports the results for the interstate banking deregulation. As shown in the subpanel B2, the coefficient on the triple interaction is statistically significant at the 1 percent level. This finding indicates that in the Sixth Federal Reserve district after the interstate banking deregulation, the increase in Chapter 13 filings to the increase in foreclosure started is significant.

Overall, the results of this subsection show that the bank deregulation increases Chapter 13 filings in the southeast region even after controlling for negative mortgage outcomes. Further, it seems that the delinquent homeowners seek protection in Chapter 13 bankruptcy to save their homes.

E. Robustness check after controlling for the state-level bankruptcy exemptions

A discussion on the effect of state-specific bankruptcy exemptions on the findings of the paper is helpful. The deregulation period of this paper is from 1980 to 1994. The major reforms in the bankruptcy process took place with the introductions of the Bankruptcy Reform Act of 1978 and the BAPCPA of 2005. Generally, the state bankruptcy exemptions change in the few years surrounding the reforms (Desai, Elliehausen & Steinbuks 2013) Therefore, it is possible that the state bankruptcy exemptions might have not changed during the 1980-94. In their specification involving state-fixed effects and state homestead exemptions, Dick and Lehnert (2010) find an insignificant effect of the state homestead exemptions. Their finding suggests that the state-fixed effects reasonably capture the state bankruptcy exemptions. In addition, Hynes, Malani and Posner (2004) show that out of the six states of the Sixth Federal Reserve district, only in GA the homestead exemptions notably changed between 1975 and 1996. As shown in our paper, the Chapter 13 filings in GA do not change significantly. Finally, the effect of state bankruptcy exemptions on the mortgage credit is an unresolved question. Chomsisengphet and Elul (2006) show that the bankruptcy exemptions are irrelevant to the mortgage lending. They argue that the exemptions are correlated with the credit scores. Failure to account for the credit score in the previous studies (Berkowitz & Hynes 1999; Lin & White 2001) has resulted in the significant impact of bankruptcy exemptions on mortgage credit. Therefore, the increase in Chapter 13 filings in the southeast region after the bank deregulation seems not to be driven by any changes in the state-level bankruptcy exemptions. Nevertheless, for completeness purpose, the following Table XIII reports the results after controlling for the state-level bankruptcy exemptions.[18]

[Insert Table XIII here]

[18] We thank Richard Hynes for sharing the bankruptcy exemptions data for the period 1980 to 1996.

Following Hynes, Malani, and Posner (2004), the variable 'Homeowner exemption' is the aggregate of homestead and personal property exemptions. It is adjusted for year 1996 dollars and is expressed in thousands. As reported, the bank deregulation effect on Chapter 13 filings in the southeast region remains significant. The effect of bankruptcy exemptions is to decrease the Chapter 13 filings, which is intuitive. As the borrower is allowed to retain her personal assets, the creditor finds the collection efforts not worth pursuing. Therefore, the borrower defaults, but does not file for bankruptcy.

V. Conclusion

This paper provides a comprehensive study on the effects of bank deregulation on personal bankruptcy filings in the US. The results show that, on average, the bank deregulation has an insignificant effect on the consumer bankruptcy. However, a closer look on the contiguous county-pairs show that the Chapter 13 filings have increased after the intrastate branching deregulation in the southeast US region covered by the Sixth Federal Reserve district. Further, the increase in mortgage defaults and foreclosures contribute to the rise in Chapter 13 filings in this region after the bank deregulation.

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Table I: Summary Statistics

Panel A: State-level data

Variable	Mean	Std. dev.	Median	25th percentile	75th Percentile	Observations
Chapter 7 bankruptcy rate	2.274	1.389	2.071	1.151	3.153	1225
Chapter 13 bankruptcy rate	0.840	0.955	0.562	0.225	1.042	1225
Interstate banking indicator (0/1)	0.742	0.438				1225
Intrastate branching indicator (0/1)	0.806	0.396				1225
Personal income growth	0.0543	0.028	0.052	0.036	0.069	1225
Personal income growth (t-1)	0.0563	0.030	0.053	0.037	0.072	1225
Unemployment rate (%) (t-1)	6.039	2.063	5.700	4.600	7.100	1225

Panel B: County-level data

Variable	Mean	Std. dev.	Median	25th percentile	75th Percentile	Observations
Chapter 7 bankruptcy rate	1.914	1.536	1.529	0.750	2.713	75970
Chapter 13 bankruptcy rate	0.746	1.290	0.284	0.075	0.810	75970
Interstate banking indicator (0/1)	0.733	0.443				76000
Intrastate branching indicator (0/1)	0.756	0.429				76000
Personal income growth	0.0546	0.067	0.050	0.028	0.076	75946
Personal income growth (t-1)	0.0570	0.070	0.052	0.028	0.080	75928
Unemployment rate (%) (t-1)	6.902	3.638	6.104	4.378	8.553	75951

Notes to Table: The Chapter 7 (Chapter 13) bankruptcy rate is the number of Chapter 7 (Chapter 13) bankruptcy filings per 1,000 population. The interstate banking indicator takes on a value of one for a given year if a state allows interstate banking in that year, and zero otherwise. The intrastate branching indicator takes on a value of one in a given year if a state permits intrastate bank branching in that year, and zero otherwise. The personal income growth is the change in per capita nominal income relative to the previous year. The unemployment rate is the ratio of people seeking a job to the total number of people in the labor market. The sample period is 1980 to 2004. The sample includes 48 states and District of Columbia, and excludes Delaware and South Dakota. In Panel B, the number of counties is 3,040.

Table II: Deregulation effect on personal bankruptcy using all state-year observations

Dependent variable:	Chapter 7 bankruptcy rate				Chapter 13 bankruptcy rate			
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Interstate banking indicator	0.053		0.041	0.107	0.020		0.032	0.024
	(0.087)		(0.086)	(0.074)	(0.086)		(0.083)	(0.079)
Intrastate branching indicator		0.061	0.055	0.117		-0.055	-0.060	-0.051
		(0.113)	(0.114)	(0.114)		(0.120)	(0.118)	(0.117)
Personal income growth				-1.610				-1.548
				(0.774)**				(0.576)***
Personal income growth (t-1)				-3.101				-2.404
				(1.072)***				(0.747)***
Unemployment rate (%)				0.088				-0.048
				(0.036)**				(0.032)
Constant	0.908	0.887	0.889	0.852	0.289	0.308	0.309	0.965
	(0.055)***	(0.065)***	(0.065)***	(0.314)***	(0.083)***	(0.094)***	(0.094)***	(0.229)***
Observations	1225	1225	1225	1225	1225	1225	1225	1225
R-squared (within)	0.86	0.86	0.86	0.87	0.50	0.50	0.50	0.51
R-squared (overall)	0.64	0.64	0.64	0.66	0.17	0.16	0.17	0.15

Notes: The table reports the results of deregulation effects on bankruptcy filings using all state-year observations. The sample excludes Delaware and South Dakota, and the sample period is 1980 to 2004. The dependent variables for the columns (i) to (iv) and (v) to (viii) are the Chapter 7 bankruptcy rate and the Chapter 13 bankruptcy rate, respectively. The variable description is in the notes to Table I. All specifications control for state and year fixed effects. The robust standard errors clustered at the state level are in parentheses below the coefficients. ** and *** indicate statistical significance at the 5% and 1% levels, respectively.

Table III: Deregulation effect on personal bankruptcy using panel data of US counties

Dependent variable:	Chapter 7 bankruptcy rate				Chapter 13 bankruptcy rate			
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Interstate banking indicator	-0.062		-0.063	-0.016	-0.014		-0.033	-0.019
	(0.087)		(0.078)	(0.085)	(0.122)		(0.126)	(0.124)
Intrastate branching indicator		-0.000	0.007	0.041		0.088	0.092	0.104
		(0.145)	(0.143)	(0.146)		(0.145)	(0.150)	(0.152)
Personal income growth				-1.315				-0.890
				(0.364)***				(0.405)**
Personal income growth (t-1)				-1.984				-0.965
				(0.447)***				(0.440)**
Unemployment rate (%)				0.030				-0.002
				(0.019)				(0.018)
Constant	0.937	0.937	0.934	1.088	0.318	0.283	0.281	0.480
	(0.069)***	(0.099)***	(0.098)***	(0.225)***	(0.090)***	(0.101)***	(0.103)***	(0.158)***
Observations	75970	75970	75970	75918	75970	75970	75970	75918
R-squared (within)	0.74	0.74	0.74	0.74	0.35	0.35	0.35	0.35
R-squared (overall)	0.45	0.45	0.45	0.45	0.11	0.11	0.11	0.11

Notes: The table reports the results of deregulation effects on bankruptcy filings using all county-year observations. The sample includes 3,040 counties excluding those of Delaware and South Dakota, and the sample period is 1980 to 2004. The dependent variables for the columns (i) to (iv) and (v) to (viii) are the Chapter 7 bankruptcy rate and the Chapter 13 bankruptcy rate, respectively. The variable description is in the notes to Table I. All specifications control for county and year fixed effects. Regressions are weighted by the average population of a county during the sample period. The robust standard errors clustered at the state level are in parentheses below the coefficients. ** and *** indicate statistical significance at the 5% and 1% levels, respectively.

Table IV: Treatment sample description

Treatment border segment (Treatment state - Neighboring control state)	Branch deregulation in treatment state	Branch deregulation in control state	Number of contiguous county- pairs in border segment
Alabama – Tennessee	1981	1985	6
Alabama – Mississippi	1981	1986	14
Alabama – Florida	1981	1988	7
Pennsylvania - West Virginia	1982	1987	6
Georgia – Florida	1983	1988	12
Massachusetts - New Hampshire	1984	1987	3
Nebraska – Missouri	1985	1990	2
Nebraska – Iowa	1985	1994	9
Tennessee - Kentucky	1985	1990	20
Tennessee – Missouri	1985	1990	2
Tennessee - Arkansas	1985	1994	3
Mississippi - Arkansas	1986	1994	5
Kansas – Missouri	1987	1990	11
Michigan – Wisconsin	1987	1990	5
North Dakota - Minnesota	1987	1993	6
West Virginia - Kentucky	1987	1990	4
Illinois – Iowa	1988	1994	9
Louisiana – Arkansas	1988	1994	8
Oklahoma - Arkansas	1988	1994	8
Texas – Arkansas	1988	1994	2
Missouri – Arkansas	1990	1994	16
Missouri – Iowa	1990	1994	12
Wisconsin - Minnesota	1990	1993	12
Wisconsin – Iowa	1990	1994	4
		Total	186

Notes: The table reports border segments of the treatment sample and the number of contiguous county-pairs on a border segment. The first state of a border segment is a treatment state and the second state is its control (comparison) state. As an example, Alabama is a treatment state and Tennessee is a control state for Alabama-Tennessee border segment. Branch deregulation year refers to the year in which a given treatment or control state removes the restrictions on intrastate bank branching. In a contiguous county-pair, the counties are separated by a state border, and one county is in a treatment state and the other is in a control state.

Table V: Descriptive statistics of bankruptcy rate for treatment and control groups

		Control group		Treatment group	
Years relative to the intrastate branching deregulation year	Statistic	Chapter 7 bankruptcy rate	Chapter 13 bankruptcy rate	Chapter 7 bankruptcy rate	Chapter 13 bankruptcy rate
-3	Mean	0.752	0.143	0.787	0.250
	Std. Dev.	0.510	0.197	0.457	0.569
	N	151	151	151	151
-2	Mean	0.848	0.137	0.901	0.248
	Std. Dev.	0.534	0.202	0.485	0.456
	N	157	157	157	157
-1	Mean	0.966	0.178	0.9657	0.286
	Std. Dev.	0.700	0.246	0.604	0.483
	N	186	186	186	186
0	Mean	1.001	0.236	1.040	0.344
	Std. Dev.	0.636	0.337	0.619	0.604
	N	185	185	185	185
1	Mean	1.101	0.254	1.153	0.438
	Std. Dev.	0.691	0.367	0.702	0.738
	N	186	186	186	186
2	Mean	1.186	0.264	1.251	0.500
	Std. Dev.	0.779	0.425	0.707	0.851
	N	186	186	186	186
3	Mean	1.175	0.284	1.356	0.660
	Std. Dev.	0.800	0.460	0.793	1.116
	N	151	151	151	151
Pre-period	Mean	0.863	0.154	0.891	0.263
	Std. Dev.	0.601	0.219	0.530	0.502
	N	494	494	494	494
Post-period	Mean	1.152	0.266	1.246	0.524
	Std. Dev.	0.755	0.416	0.734	0.903
	N	523	523	523	523

Notes: The table reports the descriptive statistics of bankruptcy filings on two dimensions -- year relative to the branching deregulation and whether a county is in the treatment or control group. *N* is the county-year observation. The Chapter 7 (Chapter 13) bankruptcy rate is the number of bankruptcy filings under the Chapter 7 (Chapter 13) bankruptcy code per 1,000 population. Pre-period consists of three, two, and one year prior to the year of intrastate bank branching deregulation in a treatment state. The post-period consists of one, two, and three years post the intrastate bank branching deregulation in a treatment state. The year zero is when the treatment state allowed the intrastate bank branching.

Table VI: Regression-based difference-in-differences using contiguous county-pairs

Dependent variable:	Chapter 7 bankruptcy rate			Chapter 13 bankruptcy rate		
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
Treatment county indicator	0.028	0.044	0.036	0.109	0.106	0.108
	(0.036)	(0.036)	(0.036)	(0.025)***	(0.025)***	(0.025)***
Post-period indicator	0.289	0.290	0.258	0.112	0.112	0.119
	(0.043)***	(0.042)***	(0.042)***	(0.021)***	(0.021)***	(0.021)***
Difference-in-Differences	0.066	0.061	0.080	0.149	0.150	0.146
	(0.058)	(0.058)	(0.057)	(0.050)***	(0.050)***	(0.050)***
Unemployment rate (%)		-0.021	-0.022		0.004	0.004
		(0.003)***	(0.003)***		(0.003)	(0.003)
Personal income growth			-0.839			0.223
			(0.224)***			(0.141)
Personal income growth (t-1)			-1.112			0.241
			(0.192)***			(0.122)**
Constant	0.863	1.042	1.184	0.154	0.122	0.089
	(0.027)***	(0.037)***	(0.044)***	(0.010)***	(0.024)***	(0.027)***
Observations	2034	2034	2034	2034	2034	2034
R-squared	0.058	0.074	0.087	0.054	0.055	0.056

Notes: The table reports the results of difference-in-differences using the treatment sample of 186 contiguous county-pairs and econometric specification of equation (5). The dependent variables for the columns (i) to (iii) and (iv) to (vi) are the Chapter 7 and the Chapter 13 bankruptcy rates, respectively. The treatment county indicator takes on a value of one for a treatment county and zero for a contiguous control county. The post-period indicator takes on a value of one for the post-deregulation period, and zero for the pre-deregulation period. The post-deregulation period contains the three years after the year in which a treatment state allowed intrastate bank branching. The pre-period consists of the three years prior to the deregulation year. The deregulation year is excluded from the analyses. The variable of interest, difference-in-differences, is the interaction term of treatment county indicator and post-period indicator. The other variable definition is in notes to Table I. The robust standard errors are below the coefficients in parentheses. ** and *** denote statistical significance at the 5% and 1% levels, respectively.

Table VII: Intrastate bank branching effect on bankruptcy filings per state

Dependent variable	Chapter 7 bankruptcy rate			Chapter 13 bankruptcy rate			Observations
State	(i)	(ii)	(iii)	(iv)	(v)	(vi)	
Alabama	0.388	0.385	0.413	-0.102	-0.132	-0.121	216
	(0.155)**	(0.157)**	(0.165)**	(0.113)	(0.117)	(0.121)	
Pennsylvania	0.305	0.286	0.167	0.052	0.049	0.051	60
	(0.156)*	(0.155)*	(0.150)	(0.024)**	(0.022)**	(0.025)**	
Georgia	0.346	0.338	0.318	0.255	0.257	0.230	136
	(0.144)**	(0.141)**	(0.136)**	(0.155)	(0.157)	(0.147)	
Massachusetts	0.050	0.022	0.023	-0.027	-0.028	-0.027	30
	(0.103)	(0.098)	(0.099)	(0.016)	(0.017)	(0.014)*	
Nebraska	-0.289	-0.263	-0.265	0.088	0.061	0.062	132
	(0.155)*	(0.162)	(0.163)	(0.052)*	(0.051)	(0.051)	
Tennessee	0.030	-0.117	-0.112	0.752	0.501	0.550	300
	(0.165)	(0.159)	(0.158)	(0.234)***	(0.210)**	(0.218)**	
Mississippi	0.605	0.658	0.658	0.101	0.182	0.190	60
	(0.273)**	(0.225)***	(0.230)***	(0.306)	(0.198)	(0.195)	
Kansas	0.163	0.187	0.157	0.020	0.030	0.042	110
	(0.197)	(0.194)	(0.190)	(0.076)	(0.075)	(0.077)	
Michigan	-0.059	-0.076	-0.105	0.007	0.008	0.011	50
	(0.228)	(0.221)	(0.291)	(0.026)	(0.025)	(0.026)	
North Dakota	0.334	0.392	0.469	-0.014	-0.004	-0.003	72
	(0.200)*	(0.189)**	(0.195)**	(0.022)	(0.020)	(0.021)	
West Virginia	0.120	0.038	-0.007	-0.045	-0.117	-0.103	40
	(0.195)	(0.189)	(0.211)	(0.107)	(0.106)	(0.114)	
Illinois	-0.222	-0.105	-0.176	0.154	0.189	0.183	108
	(0.249)	(0.242)	(0.245)	(0.116)	(0.116)	(0.121)	
Louisiana	0.021	0.017	0.028	0.641	0.617	0.628	96
	(0.189)	(0.189)	(0.183)	(0.168)***	(0.142)***	(0.139)***	
Oklahoma	-0.004	0.010	0.525	-0.188	-0.212	-0.126	96
	(0.393)	(0.391)	(0.397)	(0.116)	(0.118)*	(0.102)	
Texas	-0.145	-0.143	-0.198	0.219	0.219	0.222	24
	(0.255)	(0.261)	(0.275)	(0.116)*	(0.119)*	(0.124)*	
Missouri	0.051	-0.004	-0.064	-0.112	-0.114	-0.124	336
	(0.118)	(0.113)	(0.111)	(0.056)**	(0.055)**	(0.056)**	
Wisconsin	-0.121	-0.160	-0.161	-0.037	-0.046	-0.045	168
	(0.176)	(0.173)	(0.173)	(0.060)	(0.062)	(0.061)	

Notes: The table reports the coefficients of the difference-in-differences interaction term of equation (5) on statewide data. The robust standard errors are in parentheses below the coefficients. The dependent variables for the columns (i) to (iii) and (iv) to (vi) are the Chapter 7 and the Chapter 13 bankruptcy rates, respectively. The columns (i) and (iv) only include the treatment indicator, post-period indicator, and their interaction term measuring the difference-in-differences treatment effect. The columns (ii) and (v) add the previous year unemployment rate. The columns (iii) and (vi) further add the current and previous year personal income growth. The last column refers to number of county-year observations used for the regressions of that state. *, **, *** indicate statistical significance at the 0.10, 0.05, and 0.01 level, respectively.

Table VIII: Bank deregulation, personal bankruptcy, and the Sixth Federal Reserve district

Panel A: US state-level data

Dependent variable:	Chapter 7 bankruptcy rate				Chapter 13 bankruptcy rate			
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Intrastate branching	0.057	0.128			-0.232	-0.225		
	(0.119)	(0.123)			(0.135)*	(0.132)*		
Intrastate branching X Sixth Fed District	0.030	0.023			1.244	1.241		
	(0.211)	(0.183)			(0.376)***	(0.371)***		
Interstate banking			0.048	0.120			-0.128	-0.124
			(0.089)	(0.079)			(0.100)	(0.099)
Interstate banking X Sixth Fed District			0.044	0.087			1.423	1.404
			(0.157)	(0.168)			(0.328)***	(0.326)***
Personal income growth		-1.559		-1.438		-1.517		-1.515
		(0.780)*		(0.795)*		(0.630)**		(0.545)***
Personal income growth (t-1)		-3.081		-2.981		-2.311		-2.026
		(1.090)***		(1.088)***		(0.747)***		(0.632)***
Unemployment rate (%)		0.086		0.087		-0.048		-0.029
		(0.036)**		(0.037)**		(0.030)		(0.023)
R-squared	0.86	0.87	0.86	0.87	0.58	0.59	0.63	0.63
Observations	1225	1225	1225	1225	1225	1225	1225	1225

Panel B: US county-level data

Dependent variable:	Chapter 7 bankruptcy rate				Chapter 13 bankruptcy rate			
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Intrastate branching	-0.052	-0.007			-0.019	-0.007		
	(0.148)	(0.157)			(0.161)	(0.164)		
Intrastate branching X Sixth Fed district	0.282	0.246			0.587	0.583		
	(0.246)	(0.232)			(0.368)	(0.371)		
Interstate banking			-0.082	-0.028			-0.120	-0.105
			(0.093)	(0.101)			(0.131)	(0.131)
Interstate banking X Sixth Fed district			0.187	0.178			0.993	0.986
			(0.194)	(0.182)			(0.466)**	(0.466)**
Personal income growth		-1.289		-1.266		-0.820		-0.705
		(0.349)***		(0.398)***		(0.456)*		(0.362)*
Personal income growth (t-1)		-1.969		-1.938		-0.924		-0.762
		(0.440)***		(0.478)***		(0.467)*		(0.404)*
Unemployment rate (%)		0.029		0.030		-0.004		-0.001
		(0.018)		(0.018)*		(0.018)		(0.016)
R-squared	0.74	0.74	0.74	0.74	0.36	0.36	0.39	0.39
Observations	75970	75918	75970	75918	75970	75918	75970	75918

Notes to Table: The table reports bank deregulation effect on bankruptcy filings for the southeast region covered by the Sixth Federal Reserve district. Panel A and Panel B report the results using state-level and county-level data, respectively. For both panels, the dependent variables for columns (i) to (iv) and (v) to (viii) are the Chapter 7 and Chapter 13 bankruptcy rates, respectively. In Panel A, the indicator variable Sixth Fed district takes on a value of one for TN, LA, AL, GA, MS, or FL, and zero otherwise. The regressions control for the state and year fixed effects. In Panel B, the indicator variable Sixth Fed district takes on a value of one for all counties of AL, FL, and GA, 74 counties in eastern two-thirds of TN, 38 counties of southern LA, and 43 counties of southern MS, and zero otherwise. Regressions are weighted by the average county population for the sample period 1980 to 2004. The regressions control for county and year fixed effects. The description of other variables is in notes to Table I. In all regressions, the robust standard errors clustered at the state level are in parentheses below the coefficients. *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

Table IX: Bank Deregulation and local economic indicators in the Sixth Federal Reserve district -- state level data

Dependent variable:	Log Gini		Log Theil		Log Unemployment		Income Growth	
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Intrastate branching	-0.025		-0.047		-0.101		0.826	
	(0.008)***		(0.016)***		(0.032)***		(0.275)***	
Intrastate branching X Sixth Fed district	-0.003		-0.001		0.029		-0.042	
	(0.011)		(0.022)		(0.067)		(0.544)	
Interstate banking		-0.015		-0.029		-0.096		0.594
		(0.007)**		(0.014)**		(0.058)		(0.322)*
Interstate banking X Sixth Fed district		-0.015		-0.024		-0.063		-0.041
		(0.008)*		(0.016)		(0.057)		(0.387)
R-squared	0.30	0.29	0.37	0.36	0.67	0.67	0.61	0.60
Observations	1225	1225	1225	1225	1225	1225	1225	1225

Notes: The table reports bank deregulation effect on local economic indicators including income inequality for the southeast region covered by the Sixth Federal Reserve district. The sample includes 48 states and the District of Columbia, it excludes Delaware and South Dakota. The sample period is from 1980 to 2004. The dependent variables for columns (i) and (ii), (iii) and (iv), (v) and (vi), and (vii) and (viii) are natural logarithm of Gini coefficient, natural logarithms of Theil Index, natural logarithm of unemployment rate, and personal income growth, respectively. The Gini coefficient and Theil Index measure income inequality of a state, these data are from Beck, Levine and Levkov (2010). The indicator variable Sixth Fed district takes on a value of one for TN, LA, AL, GA, MS, or FL, and zero otherwise. The description of other variables is in notes to Table I. The regressions control for the state and year fixed effects. In all regressions, the robust standard errors clustered at the state level are in parentheses below the coefficients. *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

Table X: Bank deregulation, Sixth Federal Reserve District and Mortgage Outcomes

Panel A: For the intrastate branching deregulation

Dependent variable:	Payments due 30 days (i)	Payments due 60 days (ii)	Payments due 90 days (iii)	Foreclosure started (iv)	Foreclosure inventory (v)
Panel A1: Only intrastate branching deregulation					
Intrastate branching	0.010	-0.038	-0.183	-0.063	-0.285
	(0.136)	(0.039)	(0.080)**	(0.027)**	(0.113)**
R-squared	0.42	0.38	0.20	0.26	0.22
Observations	1225	1225	1225	1225	1225
Panel A2: With Sixth Federal Reserve district					
Intrastate branching	-0.104	-0.077	-0.265	-0.083	-0.355
	(0.136)	(0.038)*	(0.085)***	(0.028)***	(0.119)***
Intrastate branching X Sixth Fed district	0.804	0.276	0.577	0.138	0.492
	(0.243)***	(0.065)***	(0.102)***	(0.032)***	(0.129)***
R-squared	0.46	0.42	0.24	0.27	0.23
Observations	1225	1225	1225	1225	1225
Panel A3: With controls					
Intrastate branching	-0.020	-0.037	-0.176	-0.043	-0.205
	(0.128)	(0.035)	(0.086)**	(0.027)	(0.109)*
Intrastate branching X Sixth Fed district	0.796	0.271	0.568	0.134	0.478
	(0.252)***	(0.075)***	(0.137)***	(0.040)***	(0.163)***
Personal income growth	-2.735	-1.665	-3.595	-1.300	-4.447
	(0.923)***	(0.439)***	(0.972)***	(0.190)***	(0.919)***
Personal income growth (t-1)	-3.356	-1.761	-4.348	-1.550	-5.685
	(0.918)***	(0.469)***	(1.420)***	(0.439)***	(1.409)***
Unemployment rate (%)	0.088	0.034	0.069	0.043	0.171
	(0.029)***	(0.009)***	(0.021)***	(0.010)***	(0.041)***
R-squared	0.52	0.54	0.37	0.44	0.38
Observations	1225	1225	1225	1225	1225

Panel B: For the interstate banking deregulation

Dependent variable:	Payments due 30 days (i)	Payments due 60 days (ii)	Payments due 90 days (iii)	Foreclosure started (iv)	Foreclosure inventory (v)
		Panel B1: Only interstate banking deregulation			
Interstate banking	-0.007	-0.047	-0.124	-0.076	-0.333
	(0.118)	(0.032)	(0.075)	(0.036)**	(0.165)**
R-squared	0.42	0.38	0.19	0.26	0.22
Observations	1225	1225	1225	1225	1225
		Panel B2: With Sixth Federal Reserve district			
Interstate banking	-0.081	-0.072	-0.178	-0.086	-0.373
	(0.116)	(0.032)**	(0.077)**	(0.037)**	(0.169)**
Interstate banking X Sixth Fed district	0.716	0.237	0.518	0.093	0.381
	(0.239)***	(0.070)***	(0.105)***	(0.038)**	(0.137)***
R-squared	0.46	0.42	0.22	0.27	0.23
Observations	1225	1225	1225	1225	1225
		Panel B3: With controls			
Interstate banking	0.005	-0.032	-0.090	-0.045	-0.219
	(0.113)	(0.029)	(0.066)	(0.028)	(0.129)*
Interstate banking X Sixth Fed district	0.764	0.255	0.553	0.115	0.469
	(0.242)***	(0.073)***	(0.132)***	(0.041)***	(0.155)***
Personal income growth	-2.600	-1.642	-3.685	-1.307	-4.515
	(0.927)***	(0.445)***	(0.967)***	(0.205)***	(0.980)***
Personal income growth (t-1)	-3.128	-1.701	-4.295	-1.536	-5.648
	(0.889)***	(0.465)***	(1.397)***	(0.446)***	(1.469)***
Unemployment rate (%)	0.098	0.037	0.077	0.044	0.176
	(0.029)***	(0.008)***	(0.020)***	(0.010)***	(0.041)***
R-squared	0.53	0.54	0.37	0.44	0.38
Observations	1225	1225	1225	1225	1225

Notes to Table: The table reports bank deregulation effect on mortgage outcomes for the southeast region covered by the Sixth Federal Reserve district. The sample includes 48 states and the District of Columbia, it excludes Delaware and South Dakota. The sample period is from 1980 to 2004. The mortgage outcomes include delinquencies and foreclosures for non-FHA non-VA residential loans. The mortgage delinquencies are the percentages of mortgages for which the payments are due above 30 days, above 60 days, and above 90 days. The foreclosure started represents the percentage of mortgages that entered the foreclosure process during a given year. The foreclosure inventory represents the percentage of mortgages in the foreclosure process at the end of the year. Panel A and Panel B report the results for intrastate bank branching and interstate banking deregulations, respectively. Within each panels, the first sub-panel includes only the indicator variable of bank deregulation. The second sub-panel adds interaction term of bank deregulation and the Sixth Federal Reserve district indicator variables, and the third sub-panel includes local economic indicators. The indicator variable Sixth Fed district takes on a value of one for TN, LA, AL, GA, MS, or FL, and zero otherwise. The description of other variables is in notes to Table I. All the regressions control for the state and year fixed effects. The robust standard errors clustered at the state level are in parentheses below the coefficients. *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

Table XI: Bank deregulation, Sixth Federal Reserve District and Chapter 13 filings after controlling for mortgage outcomes

Dependent variable:	Chapter 13 bankruptcy rate							
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Intrastate branching	-0.161	-0.171			-0.186	-0.194		
	(0.122)	(0.119)			(0.130)	(0.124)		
Intrastate branching X Sixth Fed district	1.091	1.068			1.168	1.145		
	(0.369)***	(0.359)***			(0.385)***	(0.374)***		
Interstate banking			-0.085	-0.099			-0.079	-0.093
			(0.095)	(0.094)			(0.095)	(0.093)
Interstate banking X Sixth Fed district			1.298	1.256			1.371	1.325
			(0.327)***	(0.326)***			(0.338)***	(0.335)***
Payments due 90 days	0.265	0.305	0.242	0.267				
	(0.081)***	(0.102)***	(0.069)***	(0.088)***				
Foreclosure started					0.549	0.714	0.568	0.686
					(0.128)***	(0.199)***	(0.126)***	(0.191)***
Personal income growth		-0.421		-0.530		-0.590		-0.618
		(0.638)		(0.582)		(0.706)		(0.636)
Personal income growth (t-1)		-0.984		-0.878		-1.204		-0.972
		(0.594)		(0.586)		(0.652)*		(0.584)
Unemployment rate (%)		-0.069		-0.050		-0.079		-0.060
		(0.032)**		(0.025)**		(0.036)**		(0.028)**
R-squared	0.60	0.62	0.65	0.65	0.60	0.61	0.65	0.66
Observations	1225	1225	1225	1225	1225	1225	1225	1225

Notes: The table reports bank deregulation effect on Chapter 13 bankruptcy filings after controlling for mortgage outcomes in the southeast region covered by the Sixth Federal Reserve district. The sample includes 48 states and the District of Columbia, it excludes Delaware and South Dakota. The sample period is from 1980 to 2004. The dependent variable is the Chapter 13 bankruptcy rate. The mortgage outcomes include mortgage default and foreclosure starts for non-FHA non-VA residential loans. The variable payments due above 90 days is the percentage of mortgages for which the payments are due above 90 days. The variable foreclosure started represents the percentage of mortgages that entered the foreclosure process during a given year. The indicator variable Sixth Fed district takes on a value of one for TN, LA, AL, GA, MS, or FL, and zero otherwise. The description of other variables is in notes to Table I. The regressions control for the state and year fixed effects. In all regressions, the robust standard errors clustered at the state level are in parentheses below the coefficients. *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

Table XII: Bank deregulation, Sixth Federal Reserve district and the impact of mortgage outcome on Chapter 13 filings

Panel A: For intrastate branch deregulation

A1) Mortgage default as mortgage outcome	(i)	(ii)	(iii)
Intrastate branching	-0.335	-0.335	-0.338
	(0.150)**	(0.151)**	(0.153)**
Intrastate branching X Sixth Fed district	0.063	0.031	0.217
	(0.476)	(0.472)	(0.470)
Intrastate branching X Payments due 90 days	0.198	0.205	0.197
	(0.118)	(0.118)*	(0.125)
Sixth Fed district X Payments due 90 days	-0.214	-0.246	-0.081
	(0.511)	(0.501)	(0.492)
Intrastate branching X Sixth Fed district X Payments due 90 days	1.092	1.142	0.888
	(0.621)*	(0.610)*	(0.613)
Payments due 90 days	0.080	0.057	0.104
	(0.058)	(0.061)	(0.060)*
Personal income growth		-0.786	-0.839
		(0.623)	(0.590)
Personal income growth (t-1)		-0.465	-1.187
		(0.531)	(0.589)**
Unemployment rate (%)			-0.058
			(0.029)*
R-squared	0.62	0.62	0.63
Observations	1225	1225	1225

A2) Foreclosure start as mortgage outcome	(i)	(ii)	(iii)
Intrastate branching	-0.257	-0.257	-0.263
	(0.155)	(0.155)	(0.154)*
Intrastate branching X Sixth Fed district	-0.319	-0.327	-0.135
	(0.518)	(0.513)	(0.541)
Intrastate branching X Foreclosure started	0.359	0.373	0.354
	(0.211)*	(0.212)*	(0.238)
Sixth Fed district X Foreclosure started	-2.292	-2.370	-1.710
	(2.346)	(2.289)	(2.337)
Intrastate branching X Sixth Fed district X Foreclosure started	5.103	5.176	4.269
	(2.888)*	(2.837)*	(2.944)
Foreclosure started	0.146	0.107	0.281
	(0.210)	(0.221)	(0.265)
Personal income growth		-0.629	-0.668
		(0.686)	(0.668)
Personal income growth (t-1)		-0.244	-0.950
		(0.535)	(0.573)
Unemployment rate (%)			-0.062
			(0.033)*
R-squared	0.63	0.63	0.64
Observations	1225	1225	1225

Panel B: For interstate banking deregulation

B1) Mortgage default as mortgage outcome	(i)	(ii)	(iii)
Interstate banking	-0.231	-0.235	-0.255
	(0.148)	(0.149)	(0.155)
Interstate banking X Sixth Fed district	0.523	0.496	0.596
	(0.599)	(0.596)	(0.590)
Interstate banking X Payments due 90 days	0.154	0.163	0.170
	(0.109)	(0.109)	(0.117)
Sixth Fed district X Payments due 90 days	-0.554	-0.579	-0.433
	(0.500)	(0.491)	(0.483)
Interstate banking X Sixth Fed district X Payments due 90 days	0.920	0.962	0.779
	(0.723)	(0.713)	(0.710)
Payments due 90 days	0.116	0.091	0.119
	(0.067)*	(0.072)	(0.068)*
Personal income growth		-0.767	-0.826
		(0.567)	(0.548)
Personal income growth (t-1)		-0.484	-1.061
		(0.547)	(0.575)*
Unemployment rate (%)			-0.046
			(0.024)*
R-squared	0.65	0.66	0.66
Observations	1225	1225	1225

B2) Foreclosure start as mortgage outcome	(i)	(ii)	(iii)
Interstate banking	-0.183	-0.188	-0.215
	(0.139)	(0.141)	(0.141)
Interstate banking X Sixth Fed district	-0.197	-0.189	-0.110
	(0.460)	(0.465)	(0.461)
Interstate banking X Foreclosure started	0.425	0.453	0.498
	(0.266)	(0.270)*	(0.270)*
Sixth Fed district X Foreclosure started	-4.307	-4.294	-3.927
	(1.666)**	(1.685)**	(1.659)**
Interstate banking X Sixth Fed district X Foreclosure started	6.333	6.315	5.816
	(2.109)***	(2.115)***	(2.154)***
Foreclosure started	0.151	0.096	0.184
	(0.243)	(0.260)	(0.274)
Personal income growth		-0.678	-0.741
		(0.638)	(0.624)
Personal income growth (t-1)		-0.311	-0.913
		(0.551)	(0.530)*
Unemployment rate (%)			-0.051
			(0.027)*
R-squared	0.67	0.67	0.68
Observations	1225	1225	1225

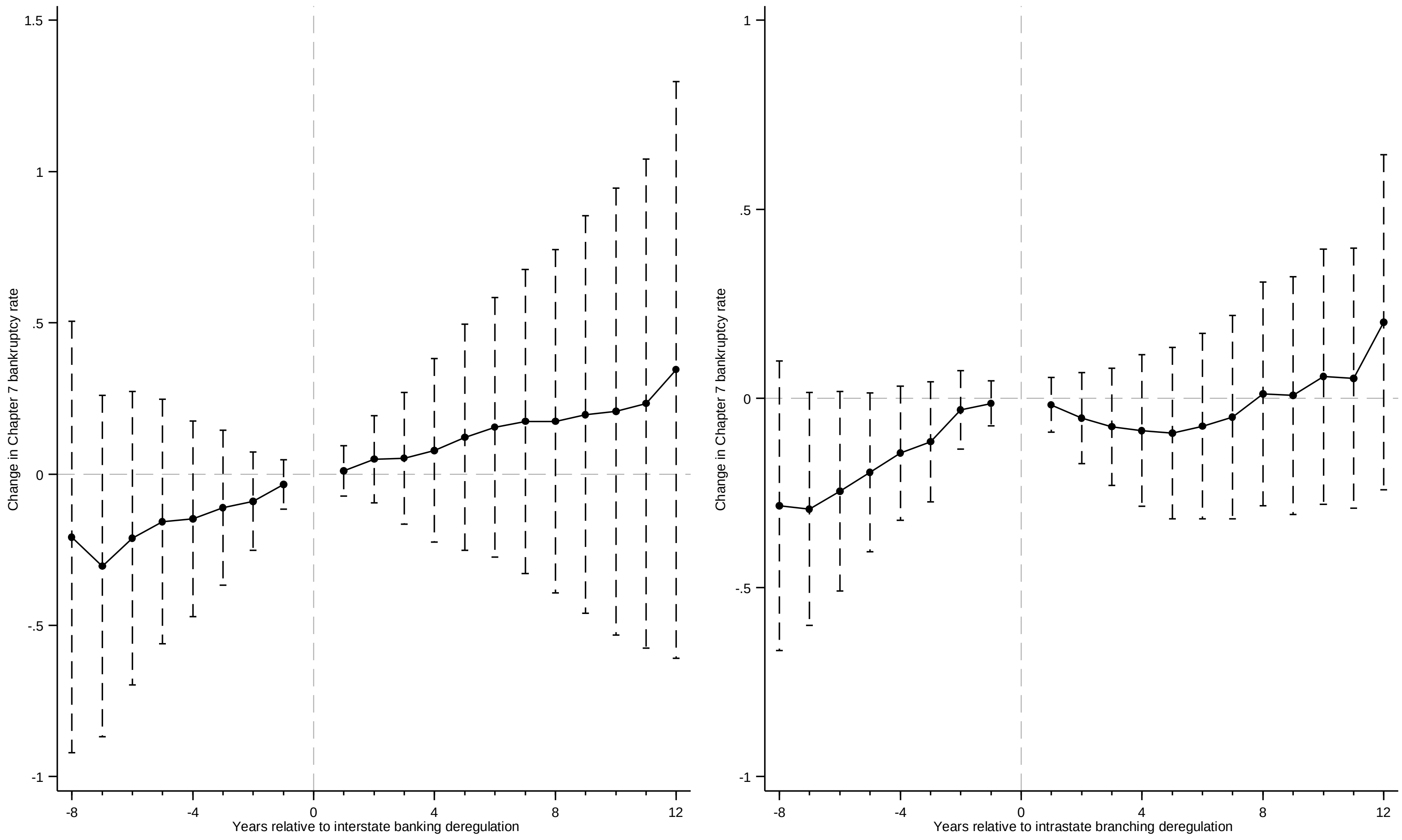
Notes to Table: The table reports the effect of mortgage outcome on Chapter 13 filings after bank deregulation in the region covered by the Sixth Federal Reserve district. Panel A and Panel B report the results for intrastate bank branching and interstate banking deregulations, respectively. The dependent variable is the Chapter 13 bankruptcy rate. The variable payments due above 90 days is the percentage of mortgages for which the payments are due above 90 days. The foreclosure started represents the percentage of mortgages that entered the foreclosure process during a given year. The indicator variable Sixth Fed district takes on a value of one for TN, LA, AL, GA, MS, or FL, and zero otherwise. The description of other variables is in notes to Table I. The regressions control for the state and year fixed effects. The robust standard errors clustered at the state level are in parentheses below the coefficients. *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

Table XIII: Bank deregulation, personal bankruptcy, and the Sixth Federal Reserve district after controlling for state-level bankruptcy exemptions

	Chapter 7 bankruptcy rate				Chapter 13 bankruptcy rate			
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Intrastate branching	0.023	0.040			-0.185	-0.188		
	(0.080)	(0.080)			(0.089)**	(0.091)**		
Intrastate branching X Sixth Fed District	0.126	0.075			0.948	0.957		
	(0.158)	(0.147)			(0.348)***	(0.346)***		
Interstate banking			0.051	0.134			-0.054	-0.060
			(0.065)	(0.059)**			(0.068)	(0.074)
Interstate banking X Sixth Fed District			0.113	0.114			1.096	1.096
			(0.113)	(0.132)			(0.308)***	(0.306)***
Personal income growth	-2.489	-1.722	-2.462	-1.719	-0.170	-0.299	-0.426	-0.483
	(0.596)***	(0.520)***	(0.595)***	(0.512)***	(0.548)	(0.540)	(0.477)	(0.496)
Personal income growth (t-1)	-4.316	-1.807	-4.305	-1.725	-0.111	-0.535	-0.304	-0.503
	(1.292)***	(0.696)**	(1.263)***	(0.646)**	(0.473)	(0.444)	(0.452)	(0.462)
Unemployment rate (%)		0.127		0.133		-0.021		-0.010
		(0.025)***		(0.026)***		(0.024)		(0.019)
Log (Homeowner exemption)	0.174	0.103	0.175	0.104	-0.186	-0.174	-0.158	-0.153
	(0.106)	(0.086)	(0.107)	(0.085)	(0.098)*	(0.094)*	(0.080)*	(0.078)*
R-squared	0.82	0.85	0.82	0.86	0.53	0.53	0.60	0.60
Observations	816	816	816	816	816	816	816	816

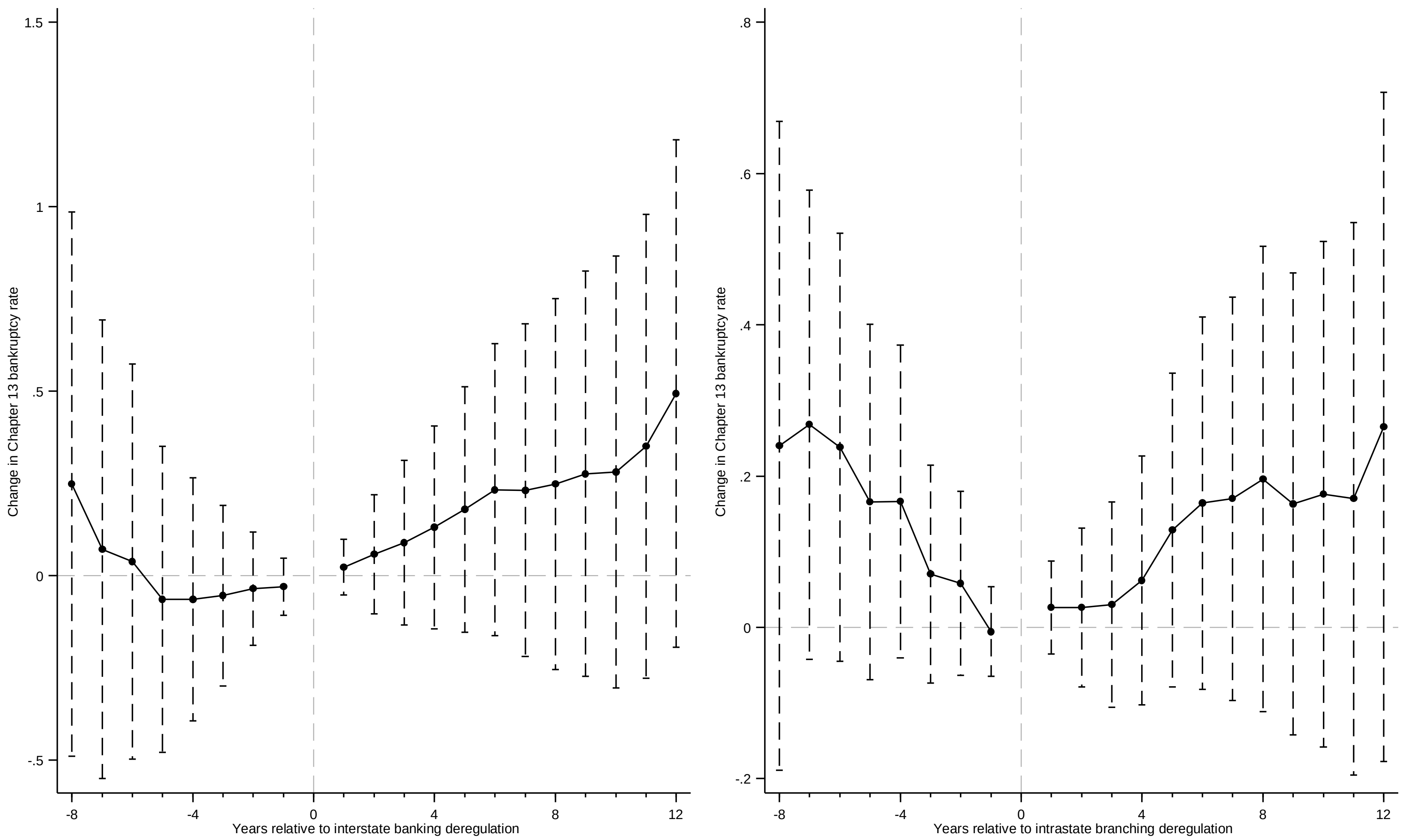
Notes: The table reports bank deregulation effect on bankruptcy filings for the southeast region covered by the Sixth Federal Reserve district after controlling for the state-level bankruptcy exemptions. The sample period is from 1980 to 1996. The dependent variables for columns (i) to (iv) and (v) to (viii) are the Chapter 7 and Chapter 13 bankruptcy rates, respectively. The indicator variable Sixth Fed district takes on a value of one for TN, LA, AL, GA, MS, or FL, and zero otherwise. Homeowner exemption is the aggregate of homestead and personal property exemptions. It is adjusted for year 1996 dollars and is expressed in thousands. These data are from Hynes, Malani, and Posner (2004). The description of other variables is in notes to Table I. The regressions control for the state and year fixed effects. In all regressions, the robust standard errors clustered at the state level are in parentheses below the coefficients. *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively.

Figure 1: Dynamic impact of deregulation on Chapter 7 bankruptcy filings using all state-level data



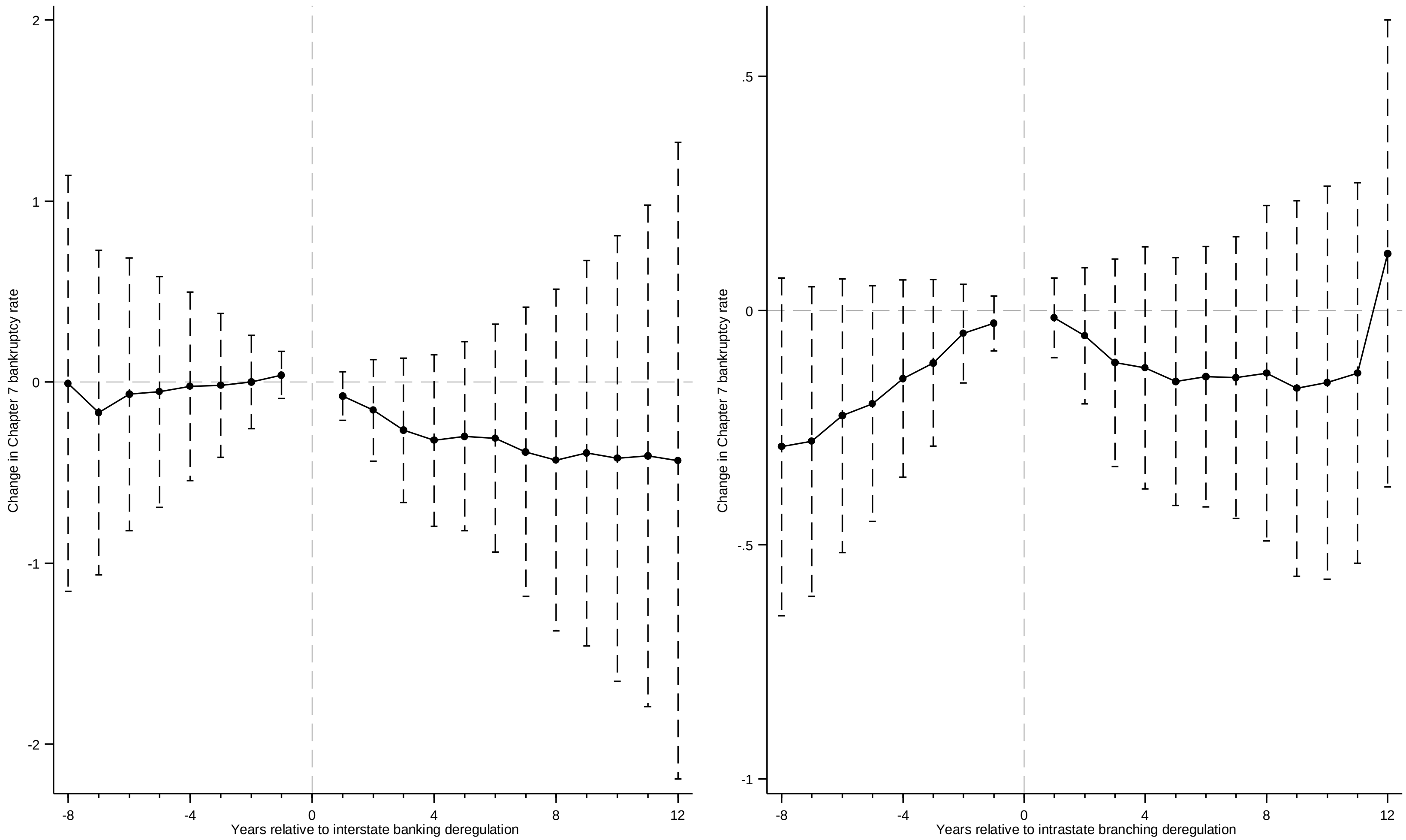
Notes: The figure plots a dynamic effect of bank deregulation on the Chapter 7 bankruptcy rate using the state-level data. The black dot is the coefficient on dummy variable $D^{\mp J}$ of the equation (2). That dummy variable takes on a value of one if the given year is J^{th} year prior to or posterior to the deregulation event year of the state s. The negative sign is for the prior year and the positive sign is for the posterior year. The dash vertical spike represents the 95% confidence interval.

Figure 2: Dynamic impact of deregulation on Chapter 13 bankruptcy filings using all state-level data



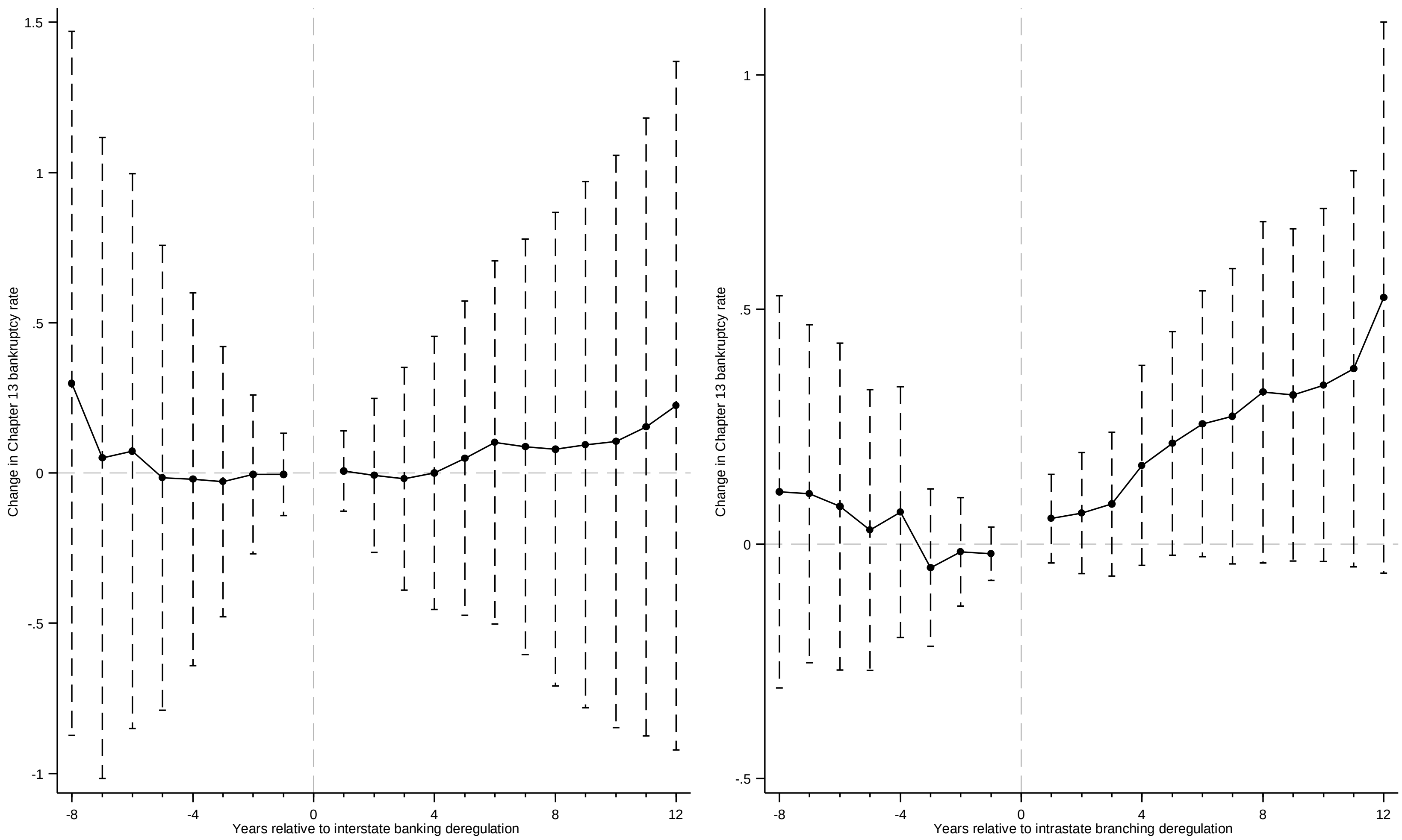
Notes: The figure plots a dynamic effect of bank deregulation on the Chapter 13 bankruptcy rate using the state-level data. The black dot is the coefficient on dummy variable $D^{\mp J}$ of the equation (2). That dummy variable takes on a value of one if the given year is J^{th} year prior to or posterior to the deregulation event year of the state s. The negative sign is for the prior year and the positive sign is for the posterior year. The dash vertical spike represents the 95% confidence interval.

Figure 3: Dynamic impact of deregulation on Chapter 7 bankruptcy filings using all county-level data



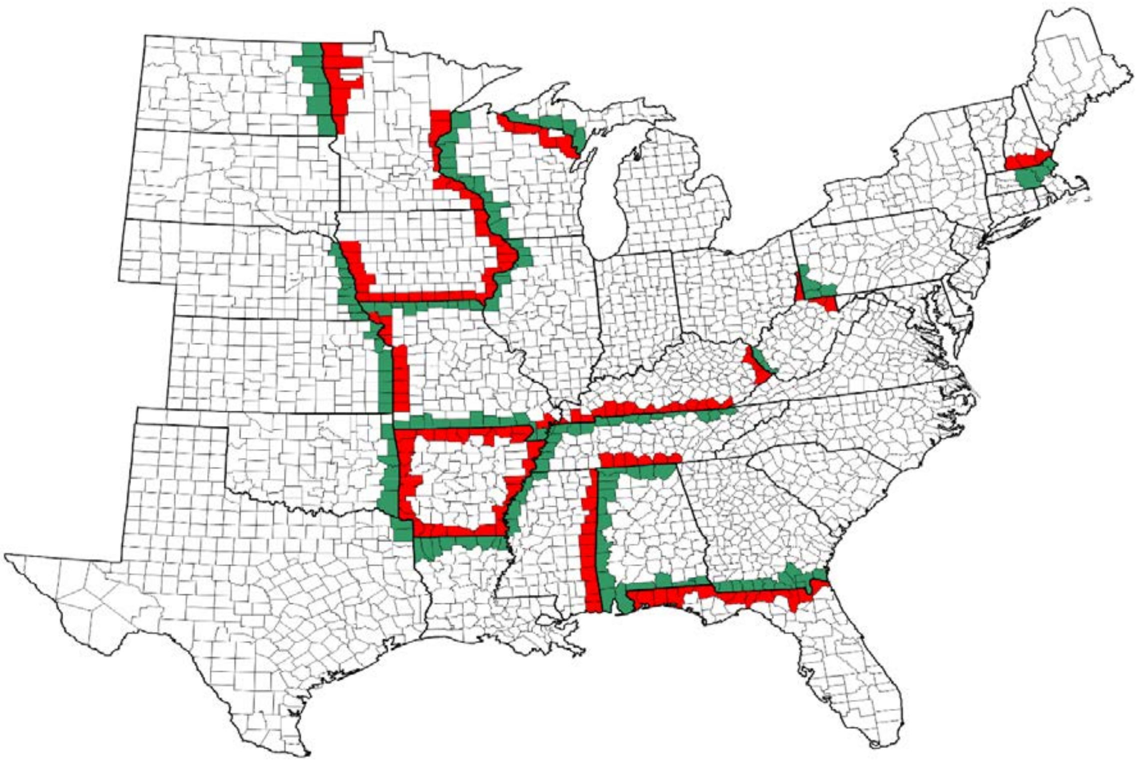
Notes: The figure plots a dynamic effect of bank deregulation on the Chapter 7 bankruptcy rate using the county-level data. The black dot is the coefficient on dummy variable $D^{\mp J}$ of the equation (4). That dummy variable takes on a value of one if the given year is J^{th} year prior to or posterior to the deregulation event year of the state s. The negative sign is for the prior year and the positive sign is for the posterior year. The dash vertical spike represents the 95% confidence interval.

Figure 4: Dynamic impact of deregulation on Chapter 13 bankruptcy filings using county-level data



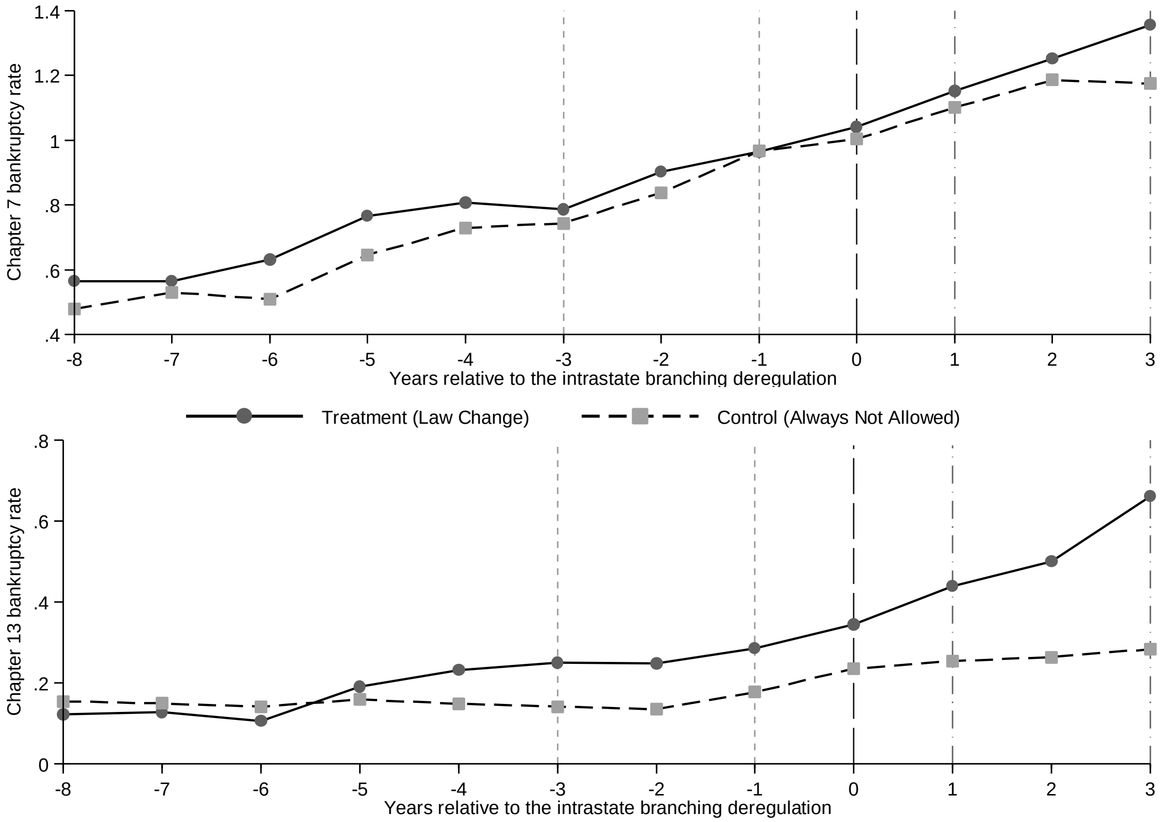
Notes: The figure plots a dynamic effect of bank deregulation on the Chapter 13 bankruptcy rate using the county-level data. The black dot is the coefficient on dummy variable $D^{\mp J}$ of the equation (4). That dummy variable takes on a value of one if the given year is J^{th} year prior to or posterior to the deregulation event year of the state s. The negative sign is for the prior year and the positive sign is for the posterior year. The dash vertical spike represents the 95% confidence interval.

Figure 5: Map of 186 contiguous-county pairs of the treatment sample



Note: The green color is for the treatment counties and the red color is for the comparison (control) counties.

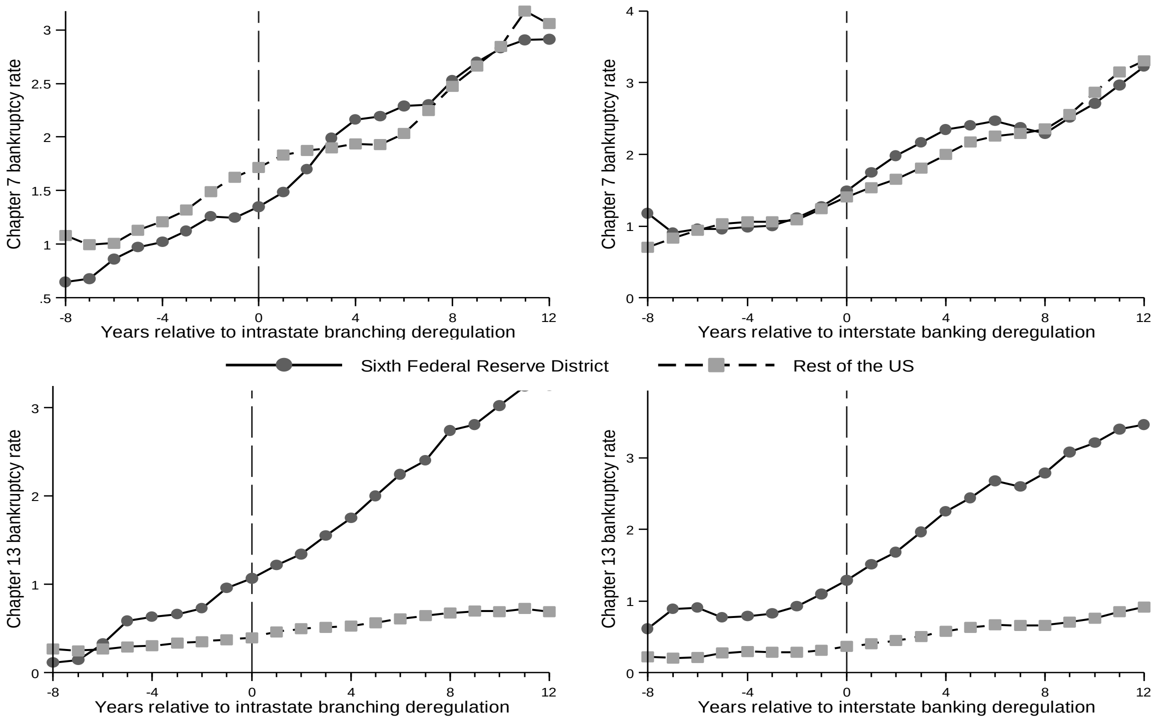
Figure 6: Trends relative to the event year for treatment and control groups



Notes: The short-dash lines are for the pre-period, the long-dash line is for the event (branch deregulation) year, and the dash-dot lines are for the post-period. These graphs plot the average Chapter 7 and 13 bankruptcy rates relative to the year in which the treatment state allowed the intrastate bank branching. The sample size is 186 counties for each treatment and control groups. The sample size for computing the average values are 151, 157, 186, 185, 186, 186, and 186 counties for years -3,-2,-1, 0, 1, 2, and 3, respectively. As we move beyond three years prior to the event year, the number of observations decline due to data limitations.

Figure 7: Bankruptcy rates in Sixth Federal Reserve district and the rest of the US

Panel A: Using the state-level data



Panel B: Using the county-level data

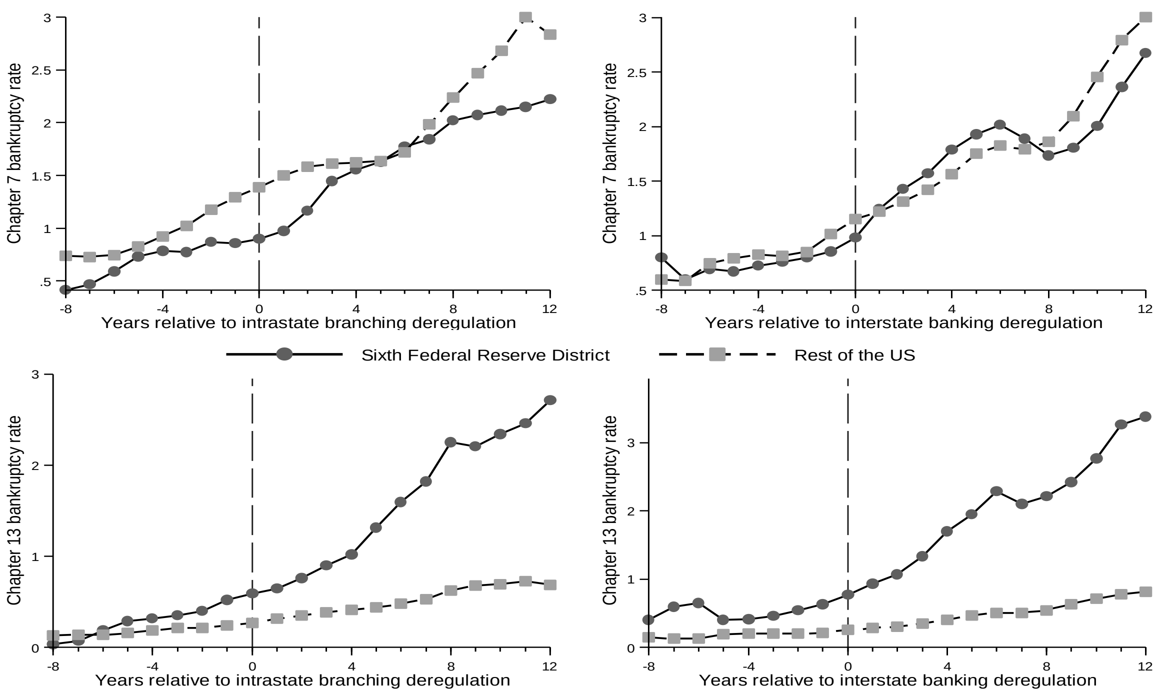
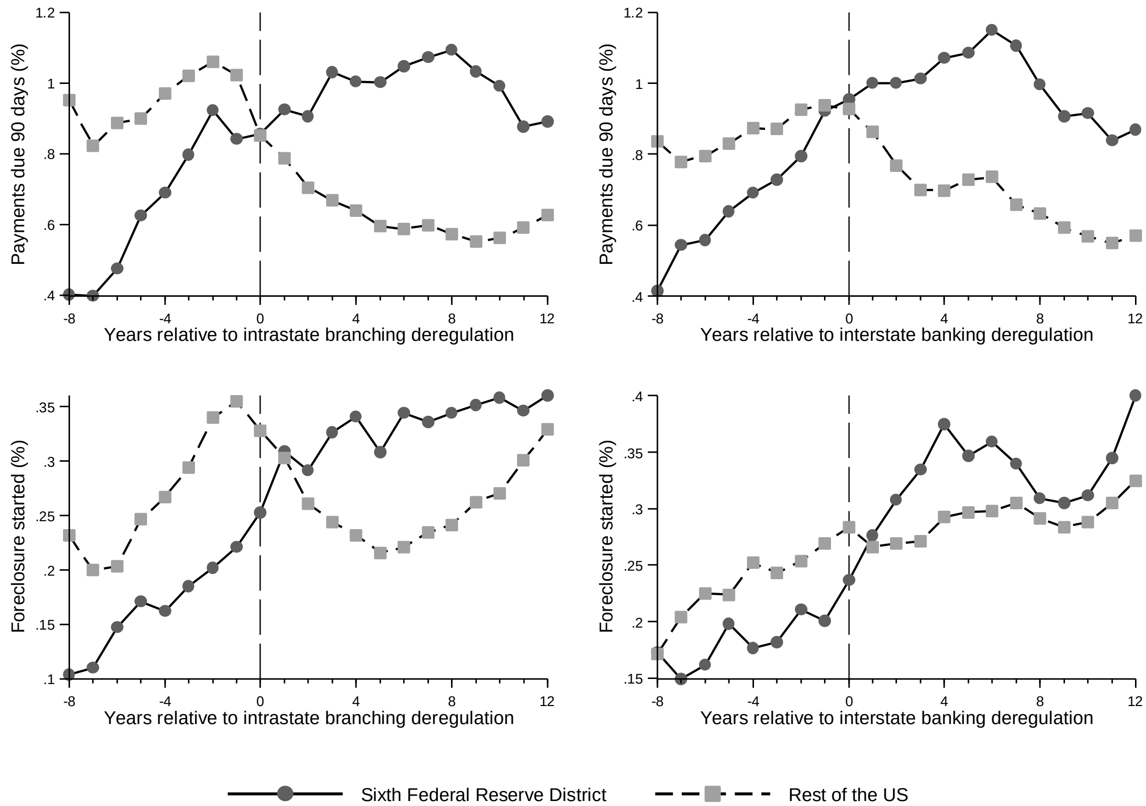


Figure 8: Mortgage outcomes in Sixth Federal Reserve district and the rest of the US



Appendices

Appendix A: Replication exercise

As documented in the Section II of the paper, the effect of interstate banking deregulation on Chapter 7 bankruptcy filings is statistically insignificant. However, Dick and Lehnert (2010), hereafter DL, show a significant effect. Therefore, the purpose of this replication exercise is to reconcile the differences in the results. We begin by replicating the results of Table III of DL using our data set. This is to make sure that inadvertently we have not made an error in our data compilation efforts. Then, we perform regressions using various combinations of their empirical model to check whether the interstate banking effect persists in all specifications. Finally, we attempt to identify the reasons of differences in the results.

Appendix Table A1 provides the descriptive statistics of the main data set used for the replication exercise. The sample period is from 1980 to 1994 and the sample size is 720 state-year observations.[20] The additional variables for the replication study are the number of top state banks, Herfindahl-Hirschman Index, deposit share of small banks, and the banks' geographic diversification from the Summary of Deposits of FDIC. We collect these data from the source files of Tewari (2014) posted on the American Economic Association website.[21] The summary statistics of Appendix Table A1 and those shown in Table II of DL suggest no major discrepancies in both data sets.

Appendix Table A2 reports the results using our data set. The regression specifications are the same as the ones in the Table III of DL. The additional reported information is the coefficient on the intercept. All specifications include state and year dummies. The results of Table A2 show

[20] Delaware, South Dakota, and the District of Columbia are excluded.
[21] We thank Ishani Tewari and the American Economic Association for posting the data on-line at https://www.aeaweb.org/articles?id=10.1257/app.6.4.175 (last accessed on: 02/06/2017).

that the interstate banking deregulation increases the Chapter 7 bankruptcy filings. Further, the results are similar to the ones reported in Table III of DL. Therefore, the first purpose of the replication exercise is accomplished.

Appendix Table A3 reports the results after modifying the specifications of the previous table. The first specification includes only the interstate banking indicator variable. The interstate banking effect is statically insignificant. Then, we include the remaining control variables in turn and assess the effect of interstate banking. The interstate banking effect becomes significant once the unemployment rate is included, as shown in column (ii). The columns (iii) and (iv) include both forms of deregulation, with and without the unemployment rate. Here also, the interstate banking effect becomes significant after the introduction of the unemployment rate. The regression specifications for columns (v) to (ix) are the same as the ones used in Table III, except the unemployment rate is excluded. As shown in Appendix Table A3, however, the effect of interstate banking persists as long as the unemployment rate is included in the specification.

A careful observation on the magnitude and statistical significance of the intercept term provides some clue about the interstate banking effect. The coefficient on the intercept decreases substantially when the unemployment rate is included in the specification. In the case of a fixed-effect cross-state and cross-time model the constant term is the average of all state-specific effects.[22] The fluctuations in the intercept coefficients suggest that there may be collinearity among the unemployment rate, the state-specific fixed effect for some of the states, the interstate banking indicator variable, and the outcome variable, the Chapter 7 bankruptcy rate. To explore this possibility, after running the regression for column (ii) of Appendix Table A3, we perform the

[22] To verify whether the results are driven by outliers in the unemployment rate, we Winsorize it at the 5 percentile high value and also use its natural logarithm. However, the results remain similar.

‘coldiag2’ command of STATA®.[23] The condition number using scaled variables is equal to 28.4, which suggests some collinearity. The Condition Indexes and Variance-Decomposition Proportions suggest the correlation among the intercept term, unemployment rate, and state effects for HI, KS, NE, NH, and ND. Out of these, four states -- HI, KS, ND, and NE -- allowed interstate banking after 1990. This regression diagnostic indicates that, by restricting the sample period till 1994, the later states are not given sufficient time to demonstrate the real impact of interstate banking on Chapter 7 filings. To test this possibility, we can either exclude the states that allowed interstate banking after 1990 while keeping the sample period from 1980 to 1994 or extend the sample period beyond 1994 while retaining all states in the analysis.

Appendix Table A4 reports the results after excluding six states -- NE, IA, ND, KS, MT, and HI-- from the analyses. These states removed the interstate banking restrictions after 1990. The sample period is from 1980 to 1994. The model specifications are the same as those in Appendix Table A2 (Table III of DL). The results show that the effect of interstate banking on the Chapter 7 bankruptcy rate is insignificant in all but two specifications. In those two specifications, the effect is statistically significant at the 10 percent level.

Appendix Table A5 reports the results based on the sample of all 48 states for the period 1980 to 1999. The main reason of ending the sample till 1999 is that the data from the source files of Tewari (2014) are up to 1999. As shown in the Table A5, the interstate banking indicator is still statistically significant, albeit with a lower impact.

To check whether the disaggregated data might have avoided the prop-up of interstate banking effect with the introduction of unemployment rate, we perform the regression analysis using the county-level data. The descriptive statistics are in Panel A of Appendix Table A6. The

[23] For more on regression diagnostics and sources of collinearity, see Belsley, Kuh and Welsch (1980).

sample period is 1980 to 1994. Since the county-level data are not available for variables based on the Summary of Deposits, the focus is only on the column (i) of the Table III of DL. Panel B of Table A6 reports the results. The regressions are weighted by the average population of a county during the sample period (1980-94). Each regression controls for time effects and county-specific fixed effects. The results show that the effect of interstate banking on Chapter 7 filings is insignificant. The effects of unemployment rate and income growth remain similar in both county- and state-level analyses.

In conclusion, the replication exercise shows that the effect of interstate banking on Chapter 7 filing is not robust to the variants in the model specifications, sample size, and data aggregation.

Appendix B: The non-parametric permutations test of counterfactuals

This Appendix provides the details of the non-parametric test. Broadly speaking, it contains three parts. The first is to identify a non-event border segment and contiguous county-pairs along that segment. The second is to prepare the sample of all counterfactuals using the fictitious event year and contiguous county-pair. The final part is to obtain the distribution of the fictitious treatment effects. Using this distribution, we can assess the deregulation effect on the actual treatment sample.

A. Identifying the non-event border-segments

Appendix Table B1 reports the non-event border segments and their number of contiguous county-pairs. A non-event border-segment is not part of treatment sample and is not a border of Montana, Wyoming, Colorado, and New Mexico. As an example, Alabama is one of the seventeen treatment states and its branch deregulation year is 1981. In the treatment sample (see Table IV), the borders of Alabama with Florida, Mississippi, and Tennessee are already considered. However, Alabama's border with Georgia was excluded because Georgia removed intrastate bank branching restrictions in 1983, which was less than two years from the event year of Alabama. In the case of the non-event border segment Pennsylvania-Maryland, the branch deregulation year for Pennsylvania is 1982. However, Maryland removed restrictions on bank branching before 1975. All twelve contiguous county-pairs on this border segment are eligible for the placebo test. Altogether, 385 contiguous county-pairs are identified from 34 non-event border segments. Appendix Figure B1 shows the non-event border segments and their contiguous county-pairs on a US map.

B. Construction of placebo sample of counterfactuals

The potential fictitious event years are between 1981 and 1992. The main reasons for selecting this period are (1) the branch deregulation period is between 1980 and 1994, and (2) we require three years for both the pre-event and post-event periods. When data are limited, we require a pre-event period of at least one year and a post-event period of at least two years. To qualify as a fictitious event year, we ensure that the fictitious event window — the period comprising the pre-event period, fictitious event year, and post-event period — does not overlap with the actual event windows of the treatment and control states. For example, considering the non-event border segment Alabama-Georgia, the years 1981 and 1983 are ineligible as fictitious event years because both are actual treatment years for Alabama and Georgia, respectively. The year 1985 is disqualified because it is part of the actual post-event period of Georgia. The year 1986 is disqualified because its prior period (i.e., 1985) is part of the post-event period for Georgia. The years 1987 to 1992 are qualified for the fictitious event years, with only differences in the length of the pre- and post- "fictitious" event periods. For the fictitious event year 1987, the pre- and post-period lengths are one and three years, respectively. For the fictitious event years 1988 and 1989, both post-periods are three years in length, however, the pre-periods are two and three years, respectively. For the fictitious event year 1992, the pre-period length is three years, but the post-period length is two years due to the sample period ending in 1994.

We identify 1,807 fictitious event year and contiguous county-pair combinations. They are part of one of the three distinct categories, depending on intrastate bank branching restrictions during the fictitious event window. In the first category, both the counties of a contiguous county-pair have restrictions on bank branching during the pre- and post-periods of the fictitious event year. As an example, consider the non-event border segment of Nebraska-Kansas and the fictitious event year is 1981. During 1980 to 1983 (fictitious event window), both states have entry barriers on intrastate bank branching. In the first category, there are 559 counterfactuals. In the second category, both counties of a contiguous county-pair allow intrastate bank branching during the fictitious event window. The fictitious event year of 1992 for the Alabama-Florida border segment is an example of this category. During the fictitious event window of 1991 to 1994, both the states permit opening of a bank branch within a state. In this category, 1,094 counterfactuals are identified. Finally, in the third category, one county of a pair restricts while the other permits the bank branching during the fictitious event window. A border segment of West Virginia-Virginia for the fictitious event year 1983 is an example of this category. West Virginia removed restrictions in 1987 and Virginia removed them in 1978. During the fictitious event window 1981-1985 surrounding fictitious event year 1983, West Virginia has regulations in place, whereas Virginia allows intrastate bank branching. From the third category, 154 counterfactuals are obtained.[24,25]

C. Parallel trend assumption for the placebo sample

For each of the 1,807 fictitious event year - contiguous county-pair combinations, pre-period average bankruptcy rates for Chapter 7 and Chapter 13 are computed by taking the average of the bankruptcy rate in the one, two, and three years prior to the fictitious event year. Similarly, the average values for the post-period is computed using bankruptcy rates for the one, two, and three years after the fictitious event year. This is done for both counties of a contiguous-county pair. Out of 1,807 pairs, data are missing in 42 pairs. Therefore, the placebo sample is of 1,765 counterfactuals.

Appendix Figure B2 plots the trends in Chapter 7 and Chapter 13 bankruptcy filings relative to the fictitious event year for the placebo sample. There is a time trend in bankruptcy filings. However, there are no notable changes in Chapter 7 filings and Chapter 13 filings after the fictitious event-year. The trends for Chapter 13 filings are parallel during the fictitious event window [-3, +3].

D. Placebo [Fictitious] treatment effect and its distribution

For each of 1,765 counterfactuals, the fictitious treatment effect is the difference-in-differences of the bankruptcy rate. It is the difference of the change in bankruptcy rate of a county

[24] While the empirical methodology of the non-parametric approach is largely based on Huang (2008), the approach differs in the identification of a combination of fictitious event-year and county-pair for the counterfactual sample. Huang (2008) constructs placebo combinations based on the 266 contiguous county-pairs for non-event borders. The fictitious event year is chosen as one of the 11 years between 1979 and 1989. This results in 266 x 11 x 2 = 5,852 possible combinations (page 691). In this paper, we make sure that there is no overlap between the fictitious event window and the actual event windows of either the treatment state or the neighboring control state.

[25] In untabulated results, 628 counterfactuals are also selected from the treatment sample. The results are similar.

and that of its contiguous county. The change is the difference in the bankruptcy rate during the post-period. The fictitious event year, here also, is not part of a pre- or post-period. Both counties of a contiguous county-pair are of the placebo sample. Therefore, they both can be interchanged for computing the fictitious treatment effect. This results in a placebo sample of 1765 x 2 = 3,530 fictitious treatment effects.

Appendix Figure B3 plots the frequency distribution of the [fictitious] treatment effect of the counterfactuals. By construction, it is symmetrical with a mean of zero. The distribution of the placebo treatment effect of Chapter 13 filings is centered to zero. This was expected based on the parallel trend in Chapter 13 filings for the placebo sample. The 90th percentile, 95th percentile, and 99th percentile of Chapter 7 bankruptcy rate are 0.627, 0.841, and 1.345, respectively. Similarly, the 90th, 95th, and 99th percentile values for Chapter 13 bankruptcy are 0.451, 0.912, and 1.878, respectively. The number 0.912, for example, informs that out of 100 random experiments using Chapter 13 bankruptcy data of a typical contiguous county-pair, in five cases we may observe the effect as large as 0.912. It also indicates that for a sample size of one contiguous county-pair, the actual treatment effect for Chapter 13 bankruptcy needs to be larger than 0.912, then only we can say that the effect is significant at the 5 percent level. While constructing the treatment effect distribution of counterfactuals, the assumption is that the contiguous county-pairs are independent of each other. Therefore, for a sample size of N contiguous county-pairs, we need to reduce the 90th, 95th, and 99th percentile cut-offs by dividing them by $\sqrt{N}$. As an example, in the case of Texas, we have two contiguous county-pairs; therefore, the actual treatment effect for the Chapter 13 bankruptcy rate needs to be above $0.645 = \frac{0.912}{\sqrt{2}}$, to be considered significant at the 5 percent level and free from data mining and data snooping biases.

E. Overall and individual state-level results of the non-parametric test

Appendix Table B2 reports the results of the non-parametric test at the treatment state and aggregate levels. As reported in the last row of the table, the treatment effect on Chapter 13 bankruptcy is 0.149 for 186 contiguous county-pairs. The cut-off value of the treatment effect at the 99th percentile level is $0.138 = \frac{1.878}{\sqrt{186}}$, when the treatment sample is of 186 contiguous county-pairs. Therefore, when evaluated using the critical values obtained from the non-parametric randomization test, the Chapter 13 bankruptcy filings, on aggregate, increase following intrastate bank branching and this increase is statistically significant at the 1 percent level. Appendix Table B2 also reports the results of our state-level analysis for Chapter 7 and Chapter 13 bankruptcy rates. The effect of intrastate bank branching on bankruptcy filings is limited to the southeast region comprising Alabama, Georgia, Mississippi, Louisiana, and Tennessee.

Appendix C: Parallel trend assumption for selected states located in southeast region

Appendix Figure C1 shows trends in bankruptcy filings for the selected treatment states with a significant effect of intrastate bank branching. Panel A of Appendix Figure C1 shows the trend in Chapter 7 filings for Georgia, and Mississippi and their control groups during the event window of [-3, +3]. The state of Alabama is excluded because the pre-event period for Alabama is only one year due to data limitation. In the case of Georgia, the average Chapter 7 bankruptcy rate is substantially increasing during the second and third year from the event year. In the case of Mississippi, the trends in Chapter 7 bankruptcy for treatment and control groups are similar during the pre-event period. However, relative to the control group, Chapter 7 bankruptcy filings increase after Mississippi enacted intrastate bank branching legislation.

Panel B of Appendix Figure C1 shows the trends in Chapter 13 filings for Tennessee and Louisiana relative to their comparison group. The pre-event trend in Chapter 13 filings for the Tennessee treatment group is similar to that of its control group. However, there is a sharp increase in Chapter 13 filings after branching deregulation in Tennessee, whereas, for the control group that trend is almost flat during the post-event period. In the case of Louisiana, after the deregulation year, its Chapter 13 bankruptcy rate gradually increased. For its contiguous control counties of Arkansas, the Chapter 13 bankruptcy rate gradually decreased after the deregulation in Louisiana. In summary, the parallel trends of selected treatment states suggest that the bankruptcy outcome paths are similar for the treatment and control groups during the pre-event period. It provides credibility to the results of parametric and non-parametric tests.

Tables to Appendices

Appendix Table A1: Descriptive statistics of state-level data

Variable	Mean	Std. dev.	Median	25th percentile	75th Percentile	Observations
Chapter 7 bankruptcy rate	1.508	0.872	1.324	0.824	2.094	720
Interstate banking indicator (0/1)	0.568	0.496				720
Intrastate branching indicator (0/1)	0.669	0.471				720
Personal income growth	0.062	0.031	0.058	0.043	0.078	720
Personal income growth (t-1)	0.065	0.032	0.063	0.045	0.085	720
Unemployment rate (%) (t-1)	6.7797	2.158	6.500	5.300	7.900	720
Number of top state banks (t-1)	22	25	12	3	32	706
HHI (t-1)	711.204	785.873	372.830	189.454	898.798	706
Deposit share of small banks (deposits < $100M)	0.253	0.213	0.183	0.069	0.408	672
Banks' geographic diversification (t-1)	3.247	4.755	1.689	1.159	3.290	705

Notes: The table reports descriptive statistics of state-year observations. The sample period is 1980 to 1994. The number of states is 48. Delaware and South Dakota are excluded. The data on deposit share of small banks are missing for 1980. HHI is an abbreviation for the Herfindahl-Hirschman Index. For some states, the data are missing for HHI, number of top state banks and geographic diversification for 1980. The HHI is rescaled after dividing it by 10,000.

Appendix Table A2: Results of replication of Dick and Lehnert (2010)

	Dependent variable: Chapter 7 bankruptcy filings per 1,000 persons							
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Interstate banking indicator	0.160	0.162	0.221	0.141	0.059	0.185	0.160	0.168
	(0.058)***	(0.057)***	(0.072)***	(0.055)**	(0.068)	(0.061)***	(0.058)***	(0.059)***
Intrastate branching indicator	0.029	0.053	0.066	0.038	0.083	0.095	0.036	0.093
	(0.073)	(0.069)	(0.065)	(0.070)	(0.068)	(0.076)	(0.072)	(0.077)
Personal income growth	-1.766	-1.608	-1.598	-1.525	-1.643	-1.449	-1.696	-1.335
	(0.527)***	(0.534)***	(0.535)***	(0.492)***	(0.486)***	(0.528)***	(0.553)***	(0.488)***
Personal income growth (t-1)	-1.487	-1.422	-1.446	-1.433	-1.501	-1.061	-1.390	-1.118
	(0.573)**	(0.574)**	(0.579)**	(0.586)**	(0.585)**	(0.527)*	(0.566)**	(0.529)**
Unemployment rate (%)	0.141	0.140	0.142	0.139	0.140	0.145	0.140	0.143
	(0.026)***	(0.025)***	(0.025)***	(0.026)***	(0.026)***	(0.027)***	(0.026)***	(0.026)***
Number of top state banks		0.003	0.003					0.001
		(0.002)	(0.002)					(0.002)
No. top banks * Interstate			-0.002					
			(0.002)					
HHI				-1.717	-2.556			-1.510
				(0.820)**	(0.721)***			(0.744)**
HHI * Interstate					1.434			
					(0.534)***			
Deposit share of small banks						1.054		0.885
						(0.482)**		(0.495)*
Banks' geographic diversification							-0.005	
							(0.003)*	
Intercept	0.453	0.404	0.365	0.583	0.616	0.672	0.496	0.848
	(0.201)**	(0.203)*	(0.200)*	(0.219)**	(0.217)***	(0.245)***	(0.206)**	(0.267)***
Observations	720	706	706	706	706	672	705	672
R^2(within)	0.84	0.84	0.84	0.85	0.85	0.85	0.84	0.85
R^2 (overall)	0.45	0.46	0.45	0.41	0.41	0.45	0.45	0.39

Notes: This table reports the results of the replication exercise of Dick and Lehnert (2010)'s Table III. For column (i), the number of observations is 720 involving 48 states for 15 years from 1980 to 1994. South Dakota and Delaware are excluded from the analyses. For some states, the data are unavailable for number of top banks, HHI, deposit share of small banks, and banks geographic diversification for year 1980. Each regression controls for time effects and state-specific fixed effects. The robust standard errors clustered at the state level are in parentheses below the coefficients. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Appendix Table A3: Unemployment rate and the significance of the interstate banking effect

	Dependent variable: Chapter 7 bankruptcy filings per 1,000 persons								
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)
Interstate banking indicator	0.031	0.155	0.032	0.155	0.065	0.097	0.046	0.069	0.077
	(0.066)	(0.059)**	(0.067)	(0.059)**	(0.062)	(0.082)	(0.058)	(0.061)	(0.060)
Intrastate branching indicator			-0.015	0.001	0.041	0.075	0.048	0.046	0.113
			(0.079)	(0.074)	(0.077)	(0.072)	(0.074)	(0.077)	(0.080)
Personal income growth					-2.511	-2.432	-2.319	-2.545	-2.468
					(0.611)***	(0.626)***	(0.598)***	(0.653)***	(0.694)***
Personal income growth (t-1)					-4.096	-4.062	-4.019	-3.995	-3.676
					(1.231)***	(1.239)***	(1.263)***	(1.230)***	(1.130)***
Unemployment rate (%)		0.155		0.155					
		(0.026)***		(0.026)***					
Number of top state banks						0.004			0.001
						(0.002)*			(0.002)
No. top banks * Interstate						-0.001			
						(0.002)			
HHI							-2.184		-1.835
							(0.803)***		(0.737)**
Deposit share of small banks									1.121
									(0.566)*
Banks' geographic diversification								-0.007	
								(0.003)*	
Intercept	0.910	0.060	0.915	0.059	1.558	1.472	1.703	1.606	1.945
	(0.041)***	(0.162)	(0.047)***	(0.164)	(0.177)***	(0.194)***	(0.191)***	(0.188)***	(0.176)***
Observations	720	720	720	720	720	706	706	705	672
R^2 (within)	0.77	0.84	0.77	0.84	0.80	0.80	0.80	0.80	0.81
R^2 (overall)	0.41	0.44	0.41	0.44	0.44	0.44	0.38	0.44	0.33

Notes: The table reports the bank deregulation effect on Chapter 7 filings with or without unemployment rate as a control variable. The number of observations is 720 involving 48 states for 15 years from 1980 to 1994. South Dakota and Delaware are excluded from the analyses. For some states, data are unavailable for variables number of top banks, HHI, deposit share of small banks, and banks geographic diversification for year 1980. Each regression controls for time effects and state-specific fixed effects. The robust standard errors clustered at the state-level are in parentheses below the coefficients. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table A4: Deregulation effect excluding those five states late in allowing interstate banking

	Dependent variable: Chapter 7 bankruptcy filings per 1,000 persons							
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Interstate banking indicator	0.119	0.118	0.164	0.096	0.006	0.125	0.123	0.105
	(0.074)	(0.072)	(0.089)*	(0.069)	(0.082)	(0.069)*	(0.074)	(0.066)
Intrastate branching indicator	0.079	0.107	0.123	0.091	0.145	0.152	0.089	0.151
	(0.083)	(0.076)	(0.070)*	(0.078)	(0.074)*	(0.085)*	(0.081)	(0.084)*
Personal income growth	-2.087	-1.907	-1.932	-1.726	-2.045	-1.292	-2.057	-1.105
	(0.748)***	(0.768)**	(0.769)**	(0.706)**	(0.685)***	(0.729)*	(0.801)**	(0.655)*
Personal income growth (t-1)	-1.635	-1.603	-1.617	-1.700	-1.779	-1.121	-1.534	-1.267
	(0.657)**	(0.637)**	(0.636)**	(0.656)**	(0.618)***	(0.618)*	(0.659)**	(0.594)**
Unemployment rate (%)	0.142	0.142	0.145	0.140	0.142	0.148	0.142	0.146
	(0.027)***	(0.027)***	(0.026)***	(0.027)***	(0.028)***	(0.028)***	(0.027)***	(0.028)***
Number of top state banks		0.002	0.003					0.001
		(0.002)	(0.003)					(0.002)
No. top banks * Interstate			-0.002					
			(0.002)					
HHI				-1.774	-2.648			-1.580
				(0.809)**	(0.711)***			(0.736)**
HHI * Interstate					1.473			
					(0.553)**			
Deposit share of small banks						1.088		0.914
						(0.498)**		(0.510)*
Banks' geographic diversification							-0.005	
							(0.003)	
Intercept	0.469	0.447	0.405	0.621	0.670	0.681	0.529	0.872
	(0.240)*	(0.231)*	(0.230)*	(0.255)**	(0.248)**	(0.267)**	(0.241)**	(0.288)***
Observations	630	617	617	617	617	588	616	588
R^2 (within)	0.85	0.85	0.85	0.85	0.86	0.86	0.85	0.86
R^2 (overall)	0.45	0.45	0.44	0.39	0.39	0.44	0.45	0.38

Notes: The table reports the results using the specifications of Appendix Table A2, however, the number of states are 42 instead of 48. It excludes six states — NE, IA, ND, KS, MT, and HI, since these states allowed interstate banking after 1990. For some states, data are unavailable for variables number of top banks, HHI, deposit share of small banks, and banks geographic diversification for year 1980. Each regression controls for time effects and state-specific fixed effects. The robust standard errors clustered at the state level are in parentheses below the coefficients. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table A5: Deregulation effect after increasing the sample period till 1999

	Dependent variable: Chapter 7 bankruptcy filings per 1,000 persons							
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Interstate banking indicator	0.130	0.140	0.201	0.115	0.015	0.168	0.130	0.155
	(0.065)*	(0.063)**	(0.079)**	(0.060)*	(0.070)	(0.065)**	(0.066)*	(0.059)**
Intrastate branching indicator	0.012	0.065	0.078	0.016	0.069	0.118	0.018	0.125
	(0.089)	(0.079)	(0.080)	(0.085)	(0.082)	(0.081)	(0.087)	(0.077)
Personal income growth	-1.496	-1.395	-1.378	-1.366	-1.410	-1.308	-1.479	-1.249
	(0.597)**	(0.611)**	(0.607)**	(0.597)**	(0.579)**	(0.716)*	(0.654)**	(0.664)*
Personal income growth (t-1)	-2.124	-2.037	-2.039	-2.025	-2.061	-1.782	-2.042	-1.771
	(0.813)**	(0.799)**	(0.798)**	(0.822)**	(0.823)**	(0.821)**	(0.812)**	(0.796)**
Unemployment rate (%)	0.133	0.131	0.133	0.133	0.129	0.130	0.133	0.129
	(0.029)***	(0.029)***	(0.028)***	(0.030)***	(0.030)***	(0.030)***	(0.029)***	(0.029)***
Number of top state banks		0.004	0.005					0.002
		(0.002)*	(0.003)*					(0.003)
No. top banks * Interstate			-0.002					
			(0.001)					
HHI				-0.911	-2.206			-0.867
				(0.805)	(0.709)***			(0.768)
HHI * Interstate					1.510			
					(0.463)***			
Deposit share of small banks						1.016		0.834
						(0.347)***		(0.370)**
Banks' geographic diversification							-0.001	
							(0.003)	
Intercept	0.540	0.445	0.404	0.610	0.682	2.474	0.566	2.599
	(0.254)**	(0.252)*	(0.246)	(0.275)**	(0.276)**	(0.241)***	(0.265)**	(0.262)***
Observations	960	946	946	946	946	912	945	912
R^2 (within)	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89
R^2 (overall)	0.62	0.62	0.62	0.60	0.60	0.61	0.62	0.59

Notes: The table reports the results using the specifications of Appendix Table A2, however, the sample period is from 1980 to 1999. For some states, data are unavailable for variables number of top banks, HHI, deposit share of small banks, and banks geographic diversification for some states for year 1980. The number of states is 48. South Dakota and Delaware are excluded from the analyses. Each regression controls for time effects and state-specific fixed effects. The robust standard errors clustered at the state level are in parentheses below the coefficients. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table A6: Deregulation effect using disaggregated data at the county level

Panel A: Descriptive statistics

Variable	Mean	Std. dev.	Median	25th percentile	75th Percentile	Observations
Chapter 7 bankruptcy rate	1.175	0.918	0.976	0.505	1.645	45560
Interstate banking indicator (0/1)	0.554	0.497				45585
Intrastate branching indicator (0/1)	0.594	0.491				45585
Personal income growth	0.0623	0.075	0.058	0.033	0.087	45539
Personal income growth (t-1)	0.0667	0.079	0.062	0.035	0.093	45522
Unemployment rate (%) (t-1)	7.814	3.915	7.072	5.095	9.714	45536

Panel B: Regression results

	Dependent variable: Chapter 7 bankruptcy filings per 1,000 persons						
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
Interstate banking indicator	-0.040	0.024		-0.034	0.029	0.003	0.047
	(0.065)	(0.077)		(0.072)	(0.078)	(0.070)	(0.076)
Intrastate branching indicator			-0.053	-0.051	-0.044	-0.007	-0.014
			(0.135)	(0.138)	(0.130)	(0.131)	(0.125)
Personal income growth						-2.356	-1.898
						(0.407)***	(0.346)***
Personal income growth (t-1)						-2.707	-1.551
						(0.377)***	(0.233)***
Unemployment rate (%)		0.088			0.088		0.076
		(0.016)***			(0.016)***		(0.016)***
Constant	0.939	0.427	0.961	0.960	0.445	1.469	0.858
	(0.067)***	(0.145)***	(0.065)***	(0.065)***	(0.146)***	(0.078)***	(0.137)***
Observations	45560	45532	45560	45560	45532	45522	45512
R-squared (within)	0.66	0.69	0.66	0.66	0.69	0.67	0.70
R-squared (overall)	0.21	0.17	0.21	0.21	0.17	0.20	0.17

Notes: The sample period is 1980 to 1994. The number of counties is 3039. The counties of Delaware, District of Columbia, and South Dakota are excluded. Regressions are weighted by the average population of a county during the sample period. Each regression controls for time effects and county-specific fixed effects. The robust standard errors clustered at the state level are in parentheses below the coefficients. *** denotes significance at the 1% level.

Appendix Table B1: Sample description of fictitious event (out-of-sample) group

Non - Event border (Treatment State - Neighboring Control State)	Branching deregulation Year in Neighboring (control) state	Number of county pairs
Alabama - Georgia	1983	18
Pennsylvania - Ohio	1979	5
Pennsylvania - New York	1976	16
Pennsylvania - New Jersey	1977	10
Pennsylvania - Maryland	Before 1975	12
Pennsylvania - Delaware	Before 1975	2
Georgia - North Carolina	Before 1975	4
Georgia - South Carolina	Before 1975	21
Georgia - Tennessee	1985	7
Massachusetts - Vermont	Before 1975	2
Massachusetts - New York	Before 1975	2
Massachusetts - Connecticut	1980	4
Massachusetts - Rhode Island	Before 1975	5
Nebraska - South Dakota	Before 1975	18
Nebraska - Kansas	1987	20
Tennessee - Virginia	1978	6
Tennessee - North Carolina	Before 1975	15
Tennessee - Mississippi	1986	8
Mississippi - Louisiana	1988	19
Kansas - Oklahoma	1988	23
Michigan - Ohio	1979	4
Michigan - Indiana	1989	8
North Dakota - South Dakota	Before 1975	10
West Virginia - Ohio	1979	18
West Virginia - Maryland	Before 1975	9
West Virginia - Virginia	1978	21
Illinois - Indiana	1989	18
Illinois - Kentucky	1990	5
Illinois - Wisconsin	1990	7
Illinois - Missouri	1990	19
Louisiana - Texas	1988	13
Oklahoma - Texas	1988	29
Oklahoma - Missouri	1990	3
Missouri - Kentucky	1990	4
	Total	385

Note: The actual branching deregulation year of the treatment state is provided in Table IV.

Appendix Table B2: Results of non-parametric test

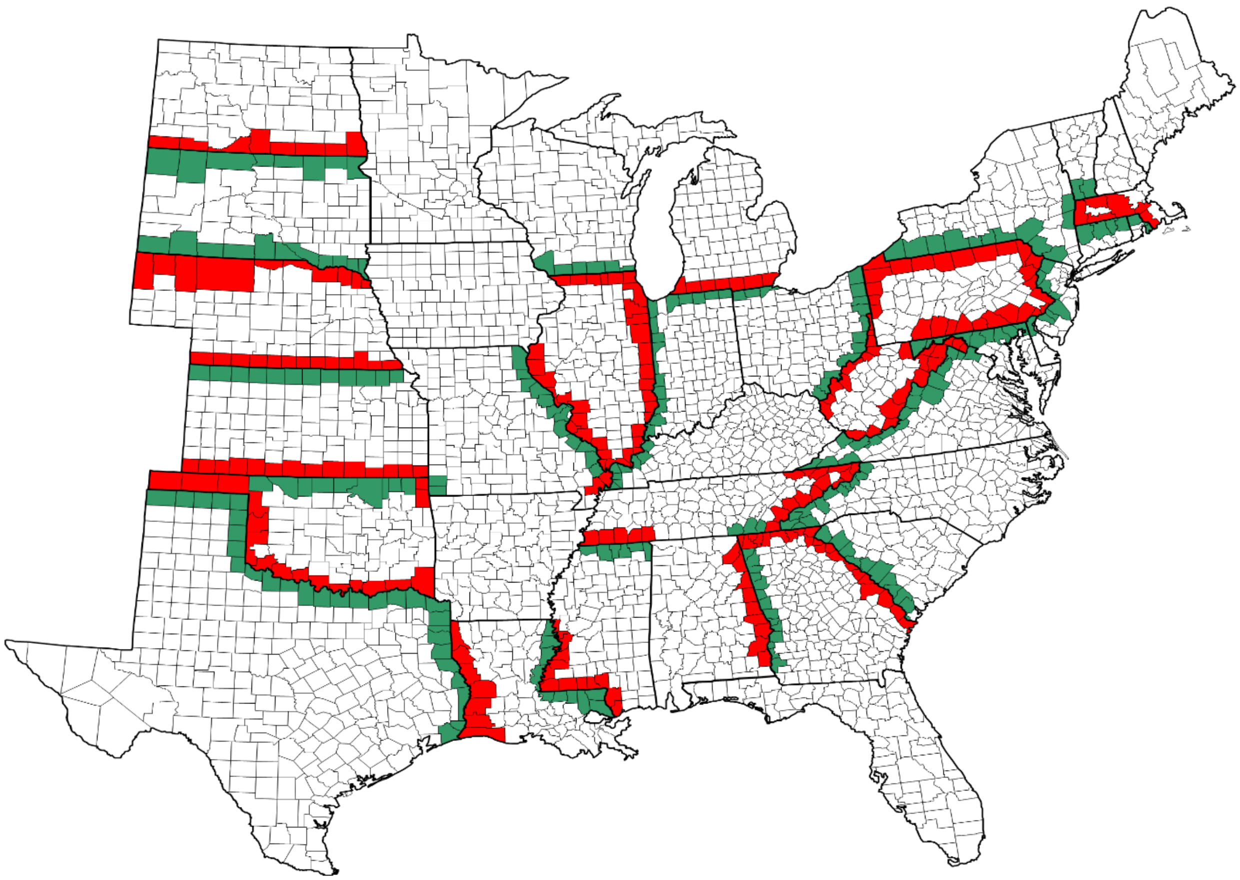
State	Chapter 7 bankruptcy rate: Average for the pre-period	Chapter 7: Treatment effect	Chapter 7: Statistical Significance	Chapter 13 bankruptcy rate: Average for the pre-period	Chapter 13: Treatment effect	Chapter 13: Statistical Significance
Alabama	0.412	0.388	1%	0.365	-0.102	10%
Pennsylvania	0.697	0.305	10%	0.034	0.052	Insignificant
Georgia	0.531	0.346	5%	0.256	0.255	10%
Massachusetts	0.329	0.050	Insignificant	0.070	-0.027	Insignificant
Nebraska	0.876	-0.289	5%	0.207	0.088	Insignificant
Tennessee	0.942	0.030	Insignificant	0.765	0.752	1%
Mississippi	1.094	0.605	1%	0.424	0.101	Insignificant
Kansas	1.268	0.163	Insignificant	0.280	0.020	Insignificant
Michigan	0.477	-0.059	Insignificant	0.013	0.007	Insignificant
North Dakota	0.696	0.334	10%	0.031	-0.014	Insignificant
West Virginia	0.536	0.120	Insignificant	0.096	-0.045	Insignificant
Illinois	1.574	-0.222	10%	0.273	0.154	10%
Louisiana	1.251	0.021	Insignificant	0.528	0.641	5%
Oklahoma	0.701	-0.004	Insignificant	0.040	-0.188	10%
Texas	0.488	-0.145	Insignificant	0.082	0.219	Insignificant
Missouri	0.949	0.051	Insignificant	0.055	-0.112	10%
Wisconsin	0.930	-0.121	Insignificant	0.063	-0.037	Insignificant
Overall	0.891	0.066	5%	0.263	0.149	1%

Appendix Table D1: Summary statistics of additional state-level variables

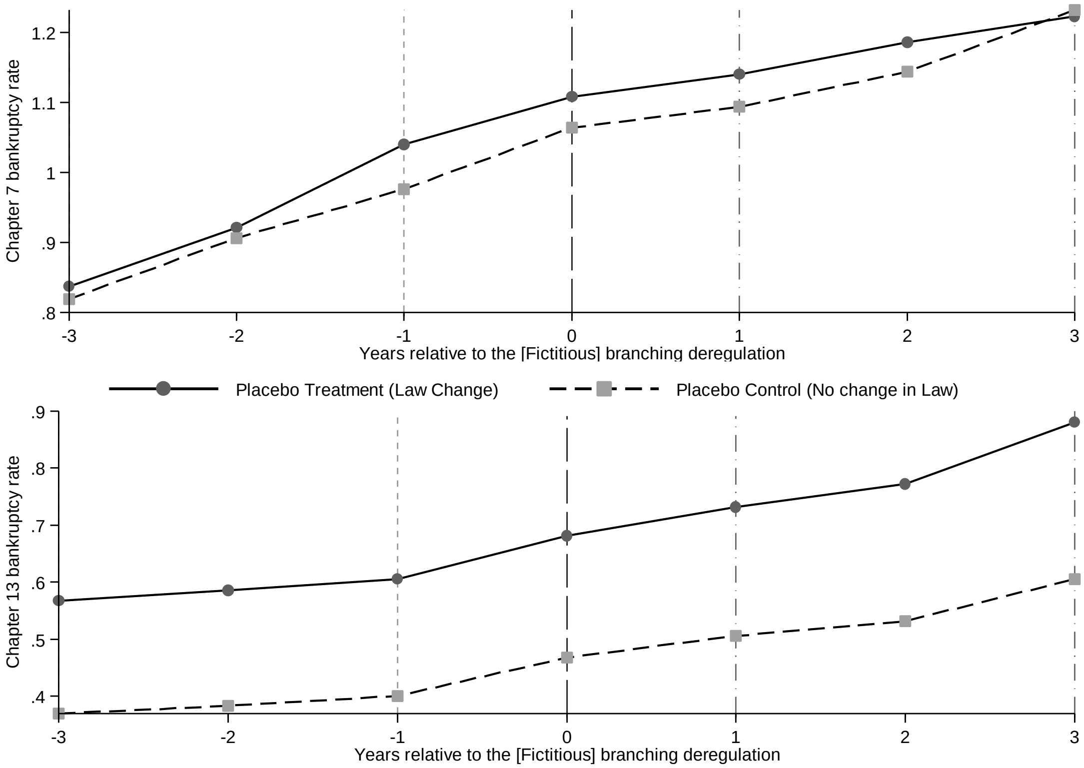
Variable	Mean	Std. dev.	Median	25th percentile	75th Percentile	Observations
Gini coefficient	0.432	0.023	0.432	0.417	0.446	1225
Theil index	0.328	0.038	0.326	0.302	0.351	1225
Payments due 30 days (%)	3.161	0.826	3.120	2.565	3.703	1225
Payments due 60 days (%)	0.757	0.235	0.733	0.588	0.910	1225
Payments due 90 days (%)	0.766	0.434	0.688	0.493	0.925	1225
Foreclosure started (%)	0.302	0.174	0.275	0.185	0.385	1225
Foreclosure inventory (%)	0.964	0.704	0.828	0.525	1.240	1225

Figures to Appendices:

Appendix Figure B1: Map of 385 contiguous county-pairs of the placebo sample

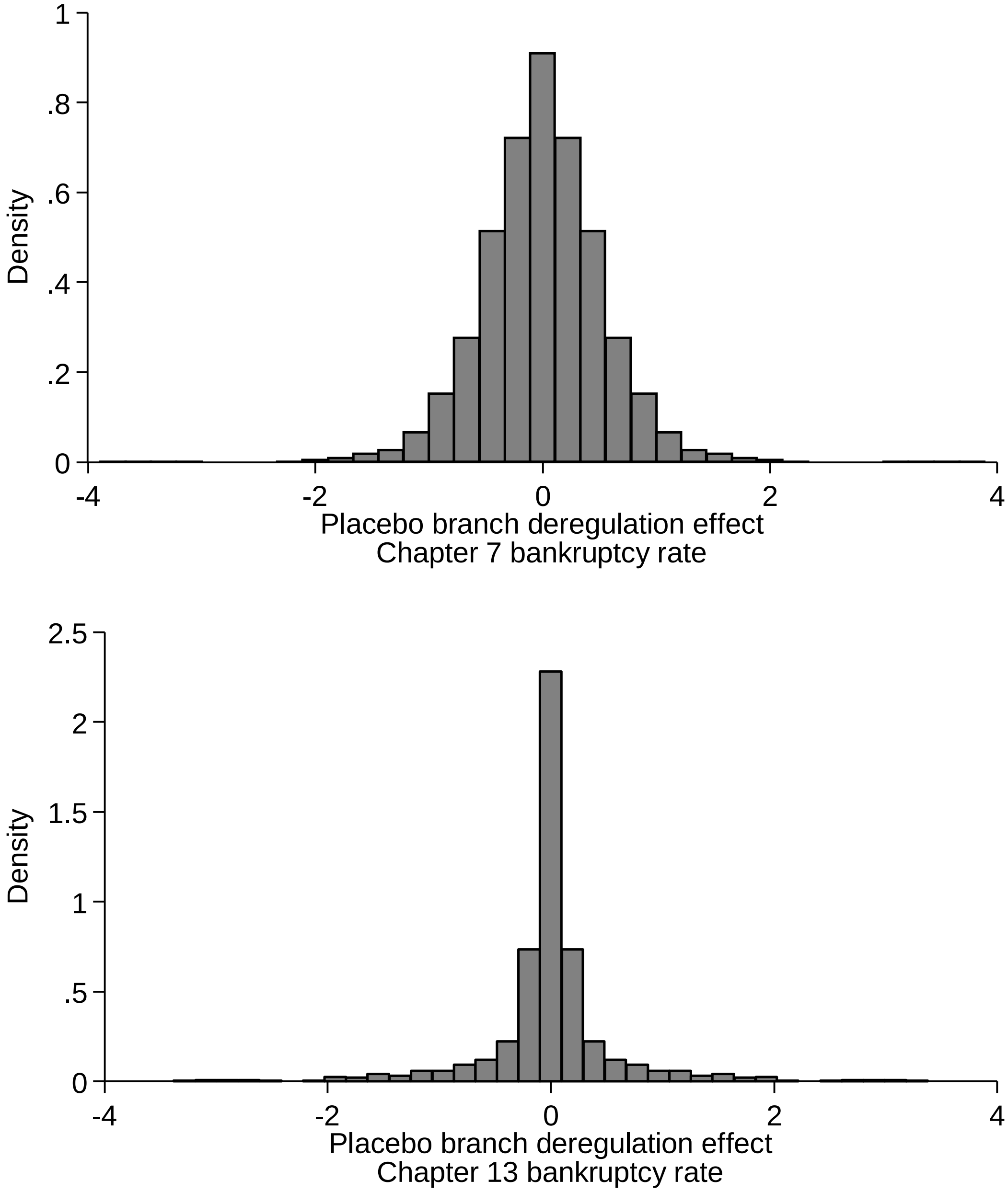


Appendix Figure B2: Trend in bankruptcy filings for the placebo sample of contiguous county-pairs



Notes: The figure is based on 1,807 county-year combinations of 385 contiguous county-pairs of the placebo sample. Year 0 represents the fictitious event year.

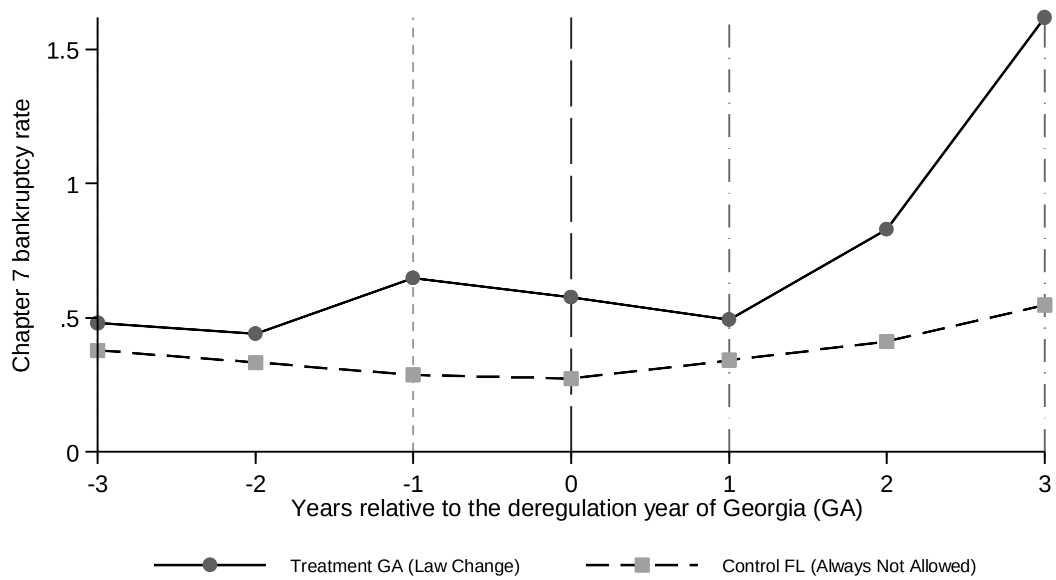
Appendix Figure B3: Empirical distribution of fictitious treatment effects

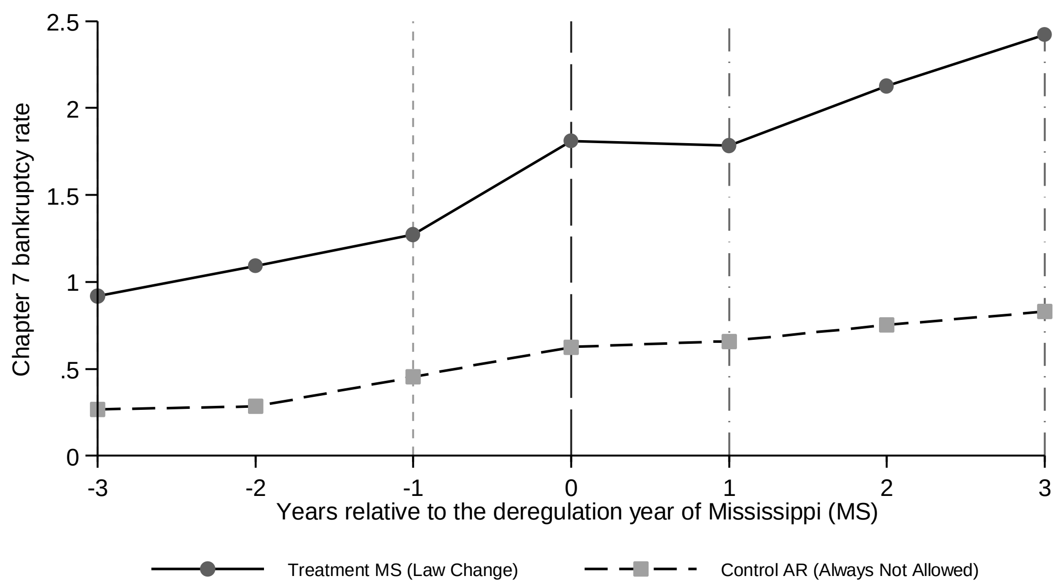


Note: The empirical distributions are based on 1765 x 2 = 3530 counterfactuals (fictitious event year - contiguous county pair) of the placebo sample.

Appendix Figure C1: State wise trend relative to the deregulation year for treatment and control states

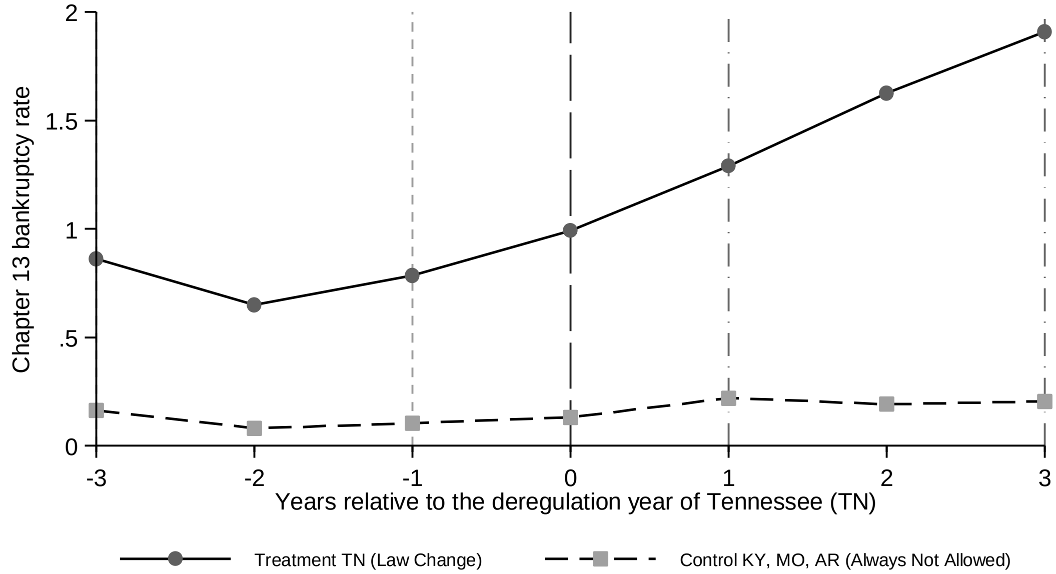
Panel A: Chapter 7 bankruptcy rate

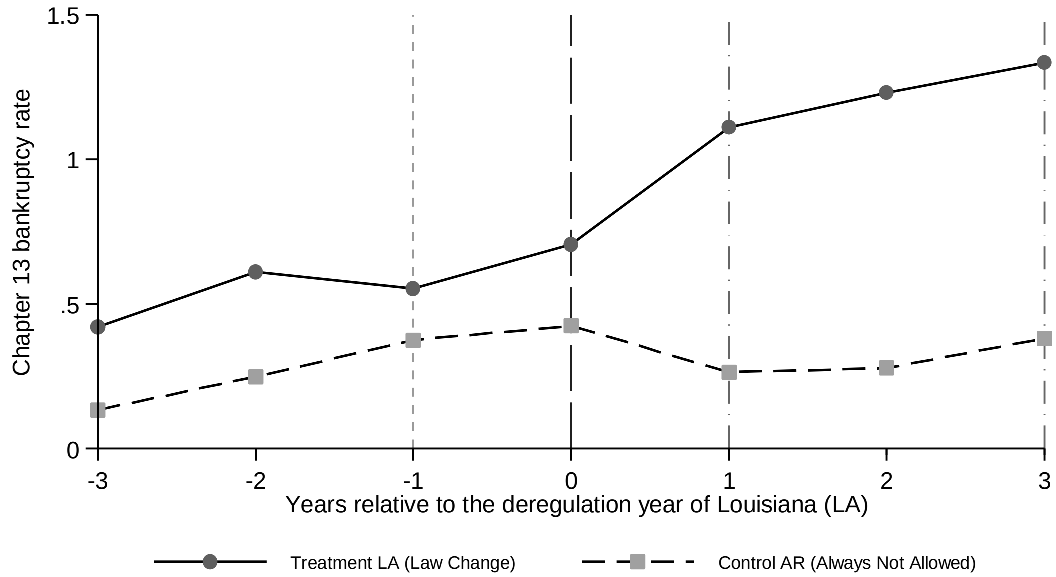




(Continued)

Panel B: Chapter 13 bankruptcy rate





Note: Year 0 is the branch deregulation (event) year when the treatment state allows intrastate bank branching.

Appendix Figure E1: Mortgage outcomes on FHA and VA loans